

# THE BURDEN OF KNOWING: POLITICAL SOPHISTICATION AND INFORMATION AVOIDANCE<sup>\*</sup>

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## Abstract

Better-skilled citizens are expected to strengthen democratic accountability, yet cognitively sophisticated citizens often polarize more on identity-relevant issues. We study how awareness of one's own motivated reasoning shapes political information acquisition. In our formal model, a citizen trading off accuracy against a directional motive decides whether to acquire a signal. Naive motivated reasoners, who do not anticipate their bias, readily acquire information. Sophisticated ones, who do, find information psychologically costly because it reduces the uncertainty that sustains desirable beliefs. We obtain four results. First, motivated reasoning alone does not produce strategic ignorance; awareness of it does. Second, for any prior, signal, and motive, this cost equals how well the signal discriminates between objective and motivated beliefs. Third, sensitivity to unfavorable news strengthens avoidance. Fourth, when beliefs guide actions, sophisticated citizens may forgo information that would improve their decisions.

*Keywords:* Bayesian updating, motivated reasoning, information acquisition, awareness, beliefs.

*Word count:* ~10300.

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# 1 Introduction

One of the central assumptions in democratic (formal) theory is that more skilled voters improve democratic performance. Informed citizens are generally better able to evaluate candidates, infer the consequences of policies, and hold political leaders accountable. For instance, informed voters rely more on policy positions than campaign spending, limiting the influence of special interests (Baron, 1994); private information enables electoral competition to aggregate dispersed knowledge (Martinelli, 2001); informed citizens are more likely to detect and punish democratic manipulation (Svolik, 2020). More broadly, democratic theory has long emphasized the importance of citizen competence and political learning for the functioning of representative government (e.g., Lupia and McCubbins, 1998; Ashworth and De Mesquita, 2014). As a result, increasing the quantity and quality of political information available to voters is typically expected to improve democratic outcomes.

A growing literature in political psychology, however, points to a striking pattern in some identity-relevant settings. Rather than uniformly making citizens more (correctly) responsive to facts, better education and cognitive skills can, in identity-relevant settings (e.g., politics), make them better at defending prior—including identity-motivated and identity-congruent—beliefs. More generally, individuals with greater cognitive sophistication often exhibit *more* polarization on identity-relevant issues because they use their cognitive abilities to defend prior beliefs (Kahan, 2013; Kahan et al., 2017). Similarly, politically sophisticated individuals generate more counterarguments against uncongenial evidence and exhibit stronger (dis)confirmation biases (Taber and Lodge, 2006; Lodge and Taber, 2013).<sup>1</sup>

Taken together, these literatures pose an interesting puzzle: why might citizens who are best equipped to understand political information (sometimes) be the least willing to seek it out? More specifically, in this paper, we ask: how can awareness of one’s own motivated reasoning *reduce* political information acquisition?

Understanding this phenomenon is crucial since many democratic institutions implicitly assume that making high-quality information publicly available is sufficient for improving political accountability and preventing democratic backsliding. Our analysis identifies an important limitation of this assumption. Even when accurate information is readily available and citizens are fully capable of understanding it, citizens who are aware of their inclination toward motivated reasoning may strategically choose not to expose themselves to evidence that threatens psychologically valuable beliefs. Our analysis suggests that improving the supply of information may therefore be insufficient if motivated citizens endogenously reduce their demand for it. In this sense, failures of democratic accountability may arise not only because citizens process information in a biased manner, but because they deliberately choose not to become informed. This point becomes particularly relevant in political environments where citizens exercise substantial control over their information exposure. Experimental and behavioral research documents partisan selection among news sources and content (Iyengar and Hahn, 2009; Mummolo, 2016), affect-

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<sup>1</sup>One of the most explicit examples of such a pattern can be found in beliefs about climate change. Rather than narrowing, the disagreement between liberals and conservatives becomes *larger* among individuals with the highest levels of scientific literacy and numeracy (Kahan et al., 2012).

driven search for candidate information (Redlawsk, 2002), partisan differences in information search that persist even under accuracy incentives (Peterson and Iyengar, 2021), and dynamic changes in real-world news consumption in response to partisan news valence (Heseltine et al., 2025).

In this paper, we consider a setting in which a decision-maker (DM) chooses whether to acquire an informative signal about an unknown state of the world. The DM values accurate beliefs but also has a directional motive to hold more desirable beliefs, so acquiring information creates a tradeoff between learning about the state and preserving scope for motivated belief formation. We distinguish between two types of DMs that engage in motivated reasoning: a naive one and a sophisticated one. The difference is that only the latter recognizes that their future beliefs will themselves be distorted by self-serving motives and anticipates this when deciding whether to become informed. We interpret this awareness (of one’s own directional motives and their consequences for future belief formation) as one dimension of political sophistication.

We obtain four main results. First, motivated reasoning alone does not generate strategic ignorance of information per se. Naive motivated reasoners may acquire information even when its direct value is slightly negative because they fail to anticipate how future motivated beliefs will affect them. Second, sophisticated motivated reasoners recognize that information reduces uncertainty and thereby narrows the scope for maintaining directionally desirable beliefs at low distortion cost, thereby inducing them to strategically avoid acquiring it.<sup>2</sup> Third, when motivated individuals additionally anticipate that future information will be unfavorable on net, incentives to avoid learning become even stronger. Finally, when beliefs subsequently guide actions, awareness of one’s own motivated reasoning creates a decision-payoff tradeoff: individuals may strategically avoid information that would objectively improve the quality of their decisions and hence their objective decision payoffs.<sup>3</sup>

This paper makes three contributions. First, it contributes to the democratic-theory literature on citizen competence (e.g., Dahl, 1989; Lupia, 1994; Delli Carpini and Keeter, 1996; Lupia and McCubbins, 1998; Campbell and Niemi, 2016; Graham and Svolik, 2020; Guess et al., 2020; Osmundsen et al., 2021; Badrinathan, 2021) by showing that, while many (cognitive) skills may be necessary for the sustainable performance of democracy, mastery of these skills alone need not be sufficient: awareness of one’s own motivated reasoning—an important dimension of cognitive sophistication—can lead citizens to anticipate how political information will constrain their identity-relevant beliefs and therefore choose not to acquire it. Second, we contribute to the formal political-science literature incorporating behavioral factors into models of political decision-making (e.g., Diermeier and Li, 2017; Lockwood, 2017; Ashworth et al., 2018; Prato and Wolton, 2018; Little, 2019; Horz, 2021, 2024; Horz and Dickson, 2024; Little et al., 2022a;

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<sup>2</sup>The Kullback-Leibler divergence driving this result distinguishes our mechanism from ones related to economic theories in which self-aware agents strategically avoid information (e.g., to preserve self-discipline (Carrillo and Mariotti, 2000), self-confidence (Bénabou and Tirole, 2002), or anticipatory utility (Brunnermeier and Parker, 2005; Köszegi, 2006)).

<sup>3</sup>Importantly, these results do not rely on the normal-linear specification. For any prior, signal structure, and directional motive, the psychological cost of information equals the divergence between the signal distributions implied by objective and motivated beliefs—information is costly exactly to the extent that it can tell the world as it is apart from the world as the citizen would like it to be—and this cost weakly increases with the informativeness of the signal and, under standard monotonicity conditions, with the strength of directional motives (Appendix B).

Nunnari and Zápál, 2025; Horz and Kocak, 2026) by extending the motivated-reasoning framework of Little (2025), showing how awareness of one’s own directional reasoning—an important factor of cognitive sophistication—can itself produce strategic ignorance. Third, we contribute to the literature on communication of (political) information (e.g., Kuklinski et al., 2000; Druckman et al., 2013; Kahan et al., 2017; Nyhan, 2021; Prato and Wolton, 2016; Merkley and Loewen, 2021; Petersen et al., 2021) by showing that policies aimed at informing citizens—especially on identity-sensitive issues—simply by increasing the availability of factual information may not be sufficient, since policy design must address not only the supply of information but also citizens’ incentives to expose themselves to it.

The remainder of the paper is organized as follows. Section 2 introduces the formal framework and characterizes both Bayesian and motivated belief updating. Section 3 studies information acquisition under motivated reasoning. We begin by introducing belief utility and then analyze information acquisition by naive and sophisticated motivated reasoners, followed by an extension involving sensitivity to favorable and unfavorable information. In Section 4 we discuss the interpretation of the model, derive testable hypotheses, and relate the theoretical predictions to the existing empirical literature. The appendices contain extensions (App. A), some general results (App. B), and proofs (App. C).

## 2 Belief Updating

Consider a continuous random variable  $\omega$  that can take values on the entire real line,  $\Omega = (-\infty, \infty)$ . Throughout the paper,  $\omega$  represents the true state of the world (i.e., any continuous state). For example, in a political context,  $\omega$  may capture the strength of democratic institutions, where larger values correspond to stronger democratic performance and smaller values correspond to greater democratic erosion.

Now consider a decision-maker (DM) who does not know the state of the world  $\omega$  and forms beliefs about it. The DM begins with an initial belief, represented by a probability density function  $f(\omega)$  such that  $\int_{-\infty}^{\infty} f(\omega) d\omega = 1$ . This function reflects the DM’s uncertainty about the state of the world. The central question is whether the DM chooses to acquire an additional signal about  $\omega$ , knowing that this signal may make beliefs more accurate but may also reduce the scope for maintaining desirable beliefs.

### 2.1 Standard Updating

Initially, let us assume that the prior belief about the state of the world  $\omega$  is represented by a normal distribution with mean  $\mu_0$  and variance  $\sigma_0^2$ , that is  $\omega \sim \mathcal{N}(\mu_0, \sigma_0^2)$ . The signal, denoted by  $s$ , which may be observed, is expressed as  $s = \omega + \epsilon$ , where  $\epsilon$  follows a normal distribution with mean 0 and variance  $\sigma_\epsilon^2$ , denoted as  $\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$  and  $\epsilon \perp \omega$ . The result of standard (Bayesian) belief updating is described in Remark 1.

**Remark 1.** Suppose the objective belief follows a normal distribution  $\omega \sim \mathcal{N}(\mu_0, \sigma_0^2)$  and there is a signal  $s = \omega + \epsilon$ , where  $\epsilon$  follows a normal distribution such that  $\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$  and  $\epsilon \perp \omega$ . Then

the posterior belief is  $\omega_s \sim \mathcal{N}(\lambda s + (1 - \lambda)\mu_0, (1 - \lambda)\sigma_0^2)$  where  $\lambda = \sigma_\epsilon^{-2}/(\sigma_\epsilon^{-2} + \sigma_0^{-2})$ .

In a standard way, the posterior mean is a weighted average of the observed signal  $s$  and the prior mean  $\mu_0$ . The weight  $\lambda$  reflects the relative precision of the signal compared to the prior belief. Specifically,  $\lambda$  increases with the precision of the signal (i.e., as  $\sigma_\epsilon^{-2}$  becomes large relative to  $\sigma_0^{-2}$ ) and decreases with the precision of the prior (i.e., as  $\sigma_0^{-2}$  becomes large relative to  $\sigma_\epsilon^{-2}$ ).

The posterior variance reflects the updated uncertainty about  $\omega$  after observing the signal. It is reduced from the prior variance  $\sigma_0^2$  based on the precision of the observed signal. The factor  $1 - \lambda$  represents the portion of the prior variance that is retained after incorporating the information from the signal. Overall, more precise signals receive a higher weight in the posterior mean and contribute more to reducing the variance.

## 2.2 Motivated Updating

Here we introduce motivated reasoning into belief updating formally.

We will denote the objective density function as  $f$  and the density function reflecting a motivated belief—i.e., the belief formed as a result of motivated reasoning—as  $\tilde{f}$ . Assume  $f_0$  represents the density of the prior belief, and  $f_s$  denotes the density of the posterior belief. This distinction allows us to analyze how beliefs are altered by cognitive biases.

Suppose there exists a *belief utility* function  $B(f, f')$  that evaluates the utility of a candidate belief  $f'$  relative to the objective belief  $f$ . The belief utility function  $B(f, f')$  is introduced to quantify how the candidate belief  $f'$  is valued relative to the objective belief  $f$ . This function can incorporate various factors such as alignment with personal (e.g., partisan) goals rather than accuracy goals (e.g., [Bolsen et al., 2014](#); [Gino et al., 2016](#); [Loewenstein and Molnar, 2018](#); [Molnar and Loewenstein, 2022](#)), self-related (i.e., political, societal, religious, etc.) identity (e.g., [Iyengar and Hahn, 2009](#); [Kahan et al., 2012](#); [Druckman et al., 2013](#); [Bullock et al., 2015](#); [Mummolo, 2016](#); [Kahan et al., 2017](#); [Druckman and McGrath, 2019](#); [Osmundsen et al., 2021](#); [Horz, 2024](#); [Zimmermann, 2020](#); [Drobner, 2022](#); [Gonzalez-Jimenez, 2022](#); [Oprea and Yuksel, 2022](#); [Thaler, 2021, 2024a](#); [Engelmann et al., 2024](#)), consistency with prior world views (cognitive consonance) (e.g., [Taber and Lodge, 2006](#); [Kahan, 2013](#); [Acharya et al., 2018](#); [Little, 2019](#); [Little et al., 2022b](#); [Little, 2025](#); [Golman et al., 2016](#); [Burro and Castagnetti, 2022](#); [Sprengholz et al., 2023](#)), or emotional satisfaction (e.g., [Redlawsk, 2002](#); [Lodge and Taber, 2013](#); [Heseltine et al., 2025](#); [Bénabou, 2015](#); [Bénabou and Tirole, 2016](#); [Stoetzer and Zimmermann, 2024](#)). The belief utility function  $B(f, f')$  provides a framework for assessing how different belief states are evaluated in the decision-making process. Throughout the analysis,  $f$  denotes the objective belief implied by the available information, whereas  $\tilde{f}$  denotes the motivated belief ultimately held by the DM.

Additionally, consider that the DM may exhibit a tendency to prioritize certain beliefs. To form a motivated belief with density  $\tilde{f}$ , the DM chooses a candidate density  $f'$  to maximize the belief utility function  $B(f, f')$ . Let  $\mathcal{F}$  be the collection of density functions  $f'$  on  $\mathbb{R}$  such that  $D_{KL}(f' \| f) < \infty$  and  $\int |v(\omega)|f'(\omega)d\omega < \infty$ .

Before studying information acquisition, we first characterize the belief that a motivated reasoner would hold conditional on a given information set. The following eq. (1) describes the belief-formation problem that determines beliefs once an information set is given. Information acquisition is introduced subsequently, after characterizing how beliefs are formed under motivated reasoning.

$$\tilde{f} = \arg \max_{f' \in \mathcal{F}} \{B(f, f')\} = \arg \max_{f' \in \mathcal{F}} \{-D_{KL}(f'|f) + v(f')\} \quad (1)$$

where  $D_{KL}(f'|f)$  is the Kullback-Leibler divergence from the candidate belief  $f'$  to the objective belief  $f$  (eq. 2), capturing the accuracy motives, and  $v(f')$  reflects the directional motives (eq. 3).

The KLD is a widely used divergence measure for quantifying the difference between probability distributions (eq. 2) (e.g., Cover, 1999; Soch et al., 2024). The KLD is nonnegative and equals zero if and only if  $f' = f$  almost everywhere; it is strictly positive whenever the two distributions differ on a set of positive measure.

$$D_{KL}(f'|f) = \int_{-\infty}^{\infty} f'(\omega) \log \left( \frac{f'(\omega)}{f(\omega)} \right) d\omega \quad (2)$$

The term  $v(f')$  denotes the directional motive for the DM (eq. 3). Little (2025) shows that this once-motivated formulation is observationally equivalent to Bayesian updating from a different prior, and that only richer formulations (twice-motivated reasoning, in which unpleasant signals are sometimes rejected) generate updating patterns with no Bayesian counterpart.

$$v(f') = \int_{-\infty}^{\infty} v(\omega) f'(\omega) d\omega \quad (3)$$

More intuitively, the DM derives satisfaction from assigning greater belief to more desirable states of the world. For example, a citizen may prefer to believe that democratic institutions remain strong, that a favored party is competent, or that a politically costly fact is less severe than available evidence might suggest. So, motivated reasoning operates by shifting beliefs toward more desirable states while still penalizing departures from the objective belief.

**Proposition 1.** *For any objective belief  $f(\cdot)$  and directional motive  $v(\cdot)$  such that  $\int_{-\infty}^{\infty} f(\omega) \exp(v(\omega)) d\omega < \infty$  and  $\int_{-\infty}^{\infty} f(\omega) \exp(v(\omega)) |v(\omega)| d\omega < \infty$ , there is a unique solution to the DM's problem stated in eq. (1):*

$$\tilde{f}(\omega) = \frac{f(\omega) \exp(v(\omega))}{\int_{-\infty}^{\infty} f(\omega') \exp(v(\omega')) d\omega'}. \quad (4)$$

The Proposition implies that motivated beliefs are obtained by reweighting the objective belief toward states that are directionally desirable. States associated with larger values of  $v(\omega)$  receive greater probability mass, while departures from the objective belief remain constrained

by the Kullback-Leibler penalty (eq. 2).

Suppose  $v(\cdot)$  is linear in  $\omega$ , such that  $v(\omega) = \alpha\omega$ , where  $\alpha \geq 0$  represents the DM's sensitivity to directional motives. In other words, higher values of  $\omega$  are psychologically more desirable for the DM. The parameter  $\alpha$  measures the strength of the directional motive: larger values of  $\alpha$  imply a stronger incentive to hold beliefs tilted toward favorable states, while  $\alpha = 0$  corresponds to standard Bayesian beliefs without directional distortion. Returning to the political illustration, suppose  $\omega$  measures the strength of democratic institutions. A citizen who is directionally motivated to believe that democracy remains healthy derives greater psychological utility from larger values of  $\omega$ , whereas a citizen committed to believing that democratic institutions have substantially deteriorated would instead prefer lower values of  $\omega$ . The parameter  $\alpha$  captures how strongly the citizen values maintaining this desirable view of democracy. When  $\alpha$  is close to zero, evidence is evaluated almost objectively. As  $\alpha$  increases, the citizen becomes more willing to distort beliefs in order to preserve the preferred view of the state of democratic institutions. The normal-linear case is particularly useful because it yields a closed-form characterization of motivated beliefs and will be used throughout the remainder of the paper.

**Corollary 1.** *If the objective belief is normal with mean  $\mu$  and variance  $\sigma^2$ , and if  $v(\cdot)$  is linear in  $\omega$ :  $v(\omega) = \alpha\omega$ , then the motivated belief is distributed normally with mean  $\mu + \alpha\sigma^2$  and variance  $\sigma^2$ .*

Applying Corollary 1 to the objective beliefs, the prior for motivated beliefs is given by

$$\tilde{\mu}_0 = \mu_0 + \alpha\sigma_0^2, \quad \tilde{\sigma}_0^2 = \sigma_0^2,$$

and the posterior for motivated beliefs is given by

$$\tilde{\mu}_s = \mu_s + \alpha\sigma_s^2, \quad \tilde{\sigma}_s^2 = \sigma_s^2,$$

or, in other terms,  $\tilde{\mu}_s = \lambda s + (1 - \lambda)\mu_0 + \alpha(1 - \lambda)\sigma_0^2$  and  $\tilde{\sigma}_s^2 = (1 - \lambda)\sigma_0^2$ , where  $\lambda = \sigma_\epsilon^{-2}/(\sigma_\epsilon^{-2} + \sigma_0^{-2})$ .

These expressions are helpful in highlighting the mechanism: motivated reasoning shifts the mean of the belief distribution in the desirable direction, but does not change the variance. Since the size of the motivated shift is proportional to the variance, uncertainty creates room for motivated beliefs. A more precise posterior therefore has two effects: it improves accuracy, but it also reduces the scope for sustaining directionally desirable beliefs. This tradeoff is what later produces strategic ignorance for self-aware motivated reasoners.

### 3 Choosing Whether to Acquire the Signal

Now we turn from belief formation to the DM's choice of whether to acquire information.

We now distinguish DMs according to how they anticipate the consequences of their directional motives when deciding whether to acquire the signal. First, suppose the DM decides to acquire the signal. In this case, the resulting utility is given by the belief utility with the posterior

belief  $f_s(\cdot)$ , and there is an overall benefit (or cost)  $b \in \mathbb{R}$  associated with acquiring the information. Now, suppose the DM chooses not to acquire the signal. Then the utility derived from the belief is given by the belief utility function  $B(f, f')$  (Little, 2025).

Consider a motivated DM holding a self-serving belief  $\tilde{f}(\cdot)$ , while their Bayesian belief is  $f(\cdot)$ . The belief utility associated with the motivated belief is then expressed as the value of the objective function evaluated at the optimal motivated belief.

**Proposition 2.** *The objective function at the optimal motivated belief is given by:*

$$\tilde{B}(f) = B(f, \tilde{f}) = \log \left( \int \exp(v(\omega)) f(\omega) d\omega \right). \quad (5)$$

The belief utility represents the logarithm of the expected utility of  $\exp(v(\omega))$  with respect to the objective density  $f(\cdot)$ . If the objective belief  $f(\cdot)$  is normally distributed with mean  $\mu$  and variance  $\sigma^2$ , and the function  $v(\cdot)$  is linear in  $\omega$ , such that  $v(\omega) = \alpha\omega$ , then  $v(\omega)$  is a normally distributed random variable with mean  $\alpha\mu$  and variance  $\alpha^2\sigma^2$ .<sup>4</sup>

Since  $\tilde{B}(f)$  is the logarithm of the moment-generating function of this random variable evaluated at  $t = 1$ , and since the moment-generating function of a normal variable is  $M(t, \mu, \sigma) = \exp\left(\mu t + \frac{t^2\sigma^2}{2}\right)$ , we have:<sup>5</sup>

$$\tilde{B}(f) = \log(M(t = 1, \alpha\mu, \alpha\sigma)) = \alpha\mu + \frac{1}{2}\alpha^2\sigma^2.$$

Intuitively, belief utility increases with higher objective beliefs, as  $\alpha > 0$ . More importantly, belief utility also increases with variance. Greater uncertainty creates more room for motivated beliefs because it allows the DM to shift beliefs toward desirable states while remaining relatively close to the objective belief. Thus, uncertainty itself becomes psychologically valuable.

The utility for retaining the prior motivated belief and selecting the optimal motivated belief is given by:

$$\tilde{B}(f_0) = \alpha\mu_0 + \frac{1}{2}\alpha^2\sigma_0^2, \quad (6)$$

where  $f_0(\cdot)$  is the prior belief with mean  $\mu_0$  and variance  $\sigma_0^2$ .

Since the DM is not sure about the mean of their posterior belief, some uncertainty is involved. If the DM acquires the signal, the resulting belief utility, conditional on the revelation  $s$ , is  $\tilde{B}(f_s) = \alpha\mu_s + \frac{1}{2}\alpha^2\sigma_s^2$  or, expressed in terms of the prior belief:

$$\tilde{B}(f_s) = \alpha(\lambda s + (1 - \lambda)\mu_0) + \frac{1}{2}\alpha^2(1 - \lambda)\sigma_0^2,$$

where  $\lambda = \sigma_\epsilon^{-2}/(\sigma_\epsilon^{-2} + \sigma_0^{-2})$ . Since this expression is linear in  $s$ , the expected belief utility across

<sup>4</sup>One can easily see that if  $W \sim \mathcal{N}(\mu, \sigma^2)$  and  $v(\omega) = \varphi W$ , then  $\varphi W \sim \mathcal{N}(\varphi\mu, \varphi^2\sigma^2)$ , since  $\mathbb{E}[v(\omega)] = \varphi\mu$  and  $\mathbb{V}[v(\omega)] = \varphi^2\sigma^2$ .

<sup>5</sup>This is a classical result (e.g., Bulmer, 1979; Kobayashi et al., 2011; Pitman, 2012; Soch et al., 2024): if a random variable  $X$  follows a normal distribution  $X \sim \mathcal{N}(\mu, \sigma^2)$ , then the moment-generating function of  $X$  is  $M_X(t) = \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right)$ .

realizations of  $s$  can be written as  $\mathbb{E}_s[\tilde{B}(f_s)] = \alpha(\lambda\mathbb{E}_s[s] + (1-\lambda)\mu_0) + \frac{1}{2}\alpha^2(1-\lambda)\sigma_0^2$ . Ultimately, the DM will acquire the signal if and only if:

$$\mathbb{E}_s[\tilde{B}(f_s)] + b \geq \tilde{B}(f_0). \quad (7)$$

This condition compares the expected value of acquiring information to the value of remaining uninformed. Information is acquired only if the overall benefit of acquisition, together with the expected change in belief utility, makes acquisition at least as valuable as remaining uninformed. This rule is the ex ante counterpart of the acceptance rule for twice-motivated reasoning in Theorem 4 in Little (2025): whereas his subject decides whether to incorporate a signal after observing it, our DM decides whether to acquire it before observing it.

One advantage of incorporating the benefit  $b$  in this simple way is that it can capture both the objective value of information ( $b^{\text{obj}}$ ) and the psychological (subjective) value of information ( $b^{\text{psy}}$ ) as a convex combination:  $b = \gamma b^{\text{obj}} + (1-\gamma)b^{\text{psy}}$ , where  $\gamma$  reflects the weight the DM assigns to each aspect. This framework allows for three types of DM: (i) a DM whose exogenous acquisition benefit is entirely objective (when  $\gamma = 1$ ), (ii) a DM who is driven entirely by the subjective value of information (e.g., affect) (when  $\gamma = 0$ ), and (iii) a DM who values both the objective and psychological aspects of information, but neither exclusively ( $\gamma \in (0, 1)$ ).

Since  $\mathbb{E}_s[s]$  plays a pivotal role in eq. (7), a crucial question arises: does the DM, when forming beliefs about the revelation of the signal, rely on the objective belief or the motivated belief? In other words, is the DM aware that their beliefs are motivated when deciding whether to acquire the information? The presence of awareness implies that the DM recognizes that directional motives systematically affect future belief formation, which constitutes one aspect of cognitive sophistication.

Now we consider two types of motivated reasoners: a sophisticated and a naive one.<sup>6</sup>

### 3.1 Sophisticated Motivated Reasoner

Let us start by assuming that the sophisticated DM is *aware* of his own directional motive and evaluates the distribution of future signals using the objective prior. Thus, we have  $\mathbb{E}_s[s] = \mu_0$ , and the expected utility of the belief with the posterior is given by  $\mathbb{E}_s[\tilde{B}(f_s)] = \alpha(\lambda\mu_0 + (1-\lambda)\mu_0) + \frac{1}{2}\alpha^2(1-\lambda)\sigma_0^2$ , that is

$$\mathbb{E}_s[\tilde{B}(f_s)] = \alpha\mu_0 + \frac{1}{2}\alpha^2(1-\lambda)\sigma_0^2, \quad (8)$$

where  $\lambda = \sigma_\epsilon^{-2}/(\sigma_\epsilon^{-2} + \sigma_0^{-2})$ .

Compared to the belief utility with the prior (eq. 6), the first term in the expected belief utility with the posterior for the self-aware motivated reasoner (eq. 8) remains the same, while the second term is reduced by a factor equal to the complement of the signal precision. This allows us to establish the following result.

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<sup>6</sup>The terminology is inspired by (O'donoghue and Rabin, 1999) and the literature on time-inconsistent preferences, where sophisticated agents correctly anticipate their future biases and naive agents do not.

**Proposition 3.** *The sophisticated DM (who is self-aware of his own directional motive) will acquire a signal  $s$  with an overall benefit  $b$  if and only if  $b \geq \frac{1}{2}\alpha^2\lambda\sigma_0^2$ .*

This threshold is strictly positive whenever  $\alpha > 0$  and  $\lambda > 0$ . Thus, in the absence of sufficiently large offsetting benefits, information is not acquired. First, the threshold for information acquisition increases with  $\alpha$ . Stronger directional motives increase the psychological value of uncertainty and therefore increase the cost of learning. As a result, more motivated individuals become less willing to acquire information. Second, the key feature of the comparative statics of the result above is that the DM does not expect information to improve the average level of future beliefs.<sup>7</sup> Since  $\mathbb{E}_s[s] = \mu_0$ , the expected posterior mean coincides with the prior mean. This follows directly from the law of iterated expectations: before observing the signal, the expected posterior mean equals the prior mean. However, new information reduces uncertainty. Since uncertainty drives belief utility, more precise information leaves less room for maintaining desirable beliefs. Thus, new information becomes psychologically costly since it reduces the uncertainty that sustains motivated beliefs.

Notably, this result does not rely on the normality assumption. For any prior, signal structure, and directional motive, a sophisticated motivated reasoner weakly dislikes information (Theorem 1 in Appendix B). In fact, the psychological cost of information equals the KLD between the signal distributions implied by the objective and by the motivated prior (Theorem 2): information is costly to the extent that it can tell the world as it is apart from the world as the DM would like it to be. In the normal-linear case, this divergence equals  $\frac{1}{2}\alpha^2\lambda\sigma_0^2$  (Corollary 5).

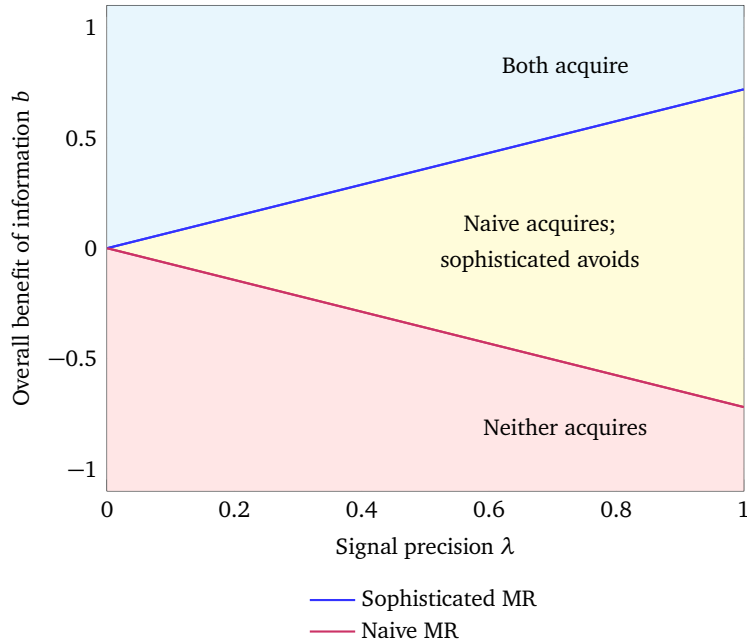


Figure 1: Information acquisition thresholds in  $(\lambda, b)$ -space by naive and sophisticated motivated reasoners.

To illustrate, consider a voter who strongly supports Donald Trump and prefers to believe

<sup>7</sup>Remarkably, this is a standard property of conditional expectations and a defining property of a martingale:  $\mathbb{E}[X_{t+1} | \mathcal{F}_t] = X_t \forall t$ , where  $\{X_t\}_{t \geq 0}$  is a sequence of random variables and  $\{\mathcal{F}_t\}_{t \geq 0}$  is a filtration.

that his conduct following the 2020 election was consistent with democratic rules. The voter can choose whether to read a detailed report describing evidence that Trump pressured state officials to change certified election results and pressed Vice President Mike Pence not to count certified electoral votes. Before reading the report, the voter retains substantial uncertainty about these events and can therefore sustain the preferred belief at relatively low distortion cost. Reading a detailed and informative report reduces this uncertainty and leaves less room for maintaining the preferred conclusion. A sophisticated motivated reasoner anticipates this consequence and may therefore choose not to read the report, whereas a naive motivated reasoner may acquire it because they fail to anticipate how the information will constrain their future motivated beliefs.

Importantly, the model does not imply that motivated individuals dislike information per se. Rather, sufficiently informative evidence can be psychologically costly because it reduces the uncertainty that sustains desirable beliefs. Figure 1 visualizes this asymmetry. The middle region is the key region of the model: for the same signal precision and the same directional motive, the naive motivated reasoner acquires information while the sophisticated motivated reasoner avoids it. Thus, the difference cannot be captured by the intensity of directional motives itself, but by awareness of what information will do to future motivated beliefs.

Overall, in the baseline belief-acquisition model, the threshold for information acquisition increases with the strength of directional motives ( $\alpha$ ), prior uncertainty ( $\sigma_0^2$ ), and signal precision ( $\lambda$ ). Consequently, highly informative signals can become less attractive because they reduce the uncertainty that allows motivated beliefs to persist.<sup>8</sup>

### 3.2 Naive Motivated Reasoner

Now suppose the DM is unaware of their directional motive when choosing whether to acquire information and therefore evaluates future signals through their motivated prior, so that  $\mathbb{E}_s[s] = \tilde{\mu}_0$ . In this case, the DM expects their belief to become more precise upon observing the new signal while remaining influenced by their original motivated belief. Since  $\tilde{\mu}_0 = \mu_0 + \alpha\sigma_0^2$ , the expected belief utility with the posterior is now  $\mathbb{E}_s[\tilde{B}(f_s)] = \alpha(\lambda\tilde{\mu}_0 + (1-\lambda)\mu_0) + \frac{1}{2}\alpha^2(1-\lambda)\sigma_0^2$ , that is

$$\mathbb{E}_s[\tilde{B}(f_s)] = \alpha\mu_0 + \frac{1}{2}\alpha^2(1+\lambda)\sigma_0^2. \quad (9)$$

where  $\lambda = \sigma_\epsilon^{-2}/(\sigma_\epsilon^{-2} + \sigma_0^{-2})$ .

Compared to the belief utility with the prior (eq. 6), the expected belief utility with the posterior for the unaware motivated reasoner (eq. 9) is larger by  $\frac{1}{2}\alpha^2\lambda\sigma_0^2$ , strictly so whenever  $\alpha > 0$  and  $\lambda > 0$ . Based on this, we formalize the following result.

**Proposition 4.** *The naive DM (who is unaware of his own directional motive) will acquire a signal  $s$  with overall benefit  $b$  if and only if  $b \geq -\frac{1}{2}\alpha^2\lambda\sigma_0^2$ .*

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<sup>8</sup>Notably, these comparative statics also extend beyond the normal-linear case. Any Blackwell more informative signal is weakly more psychologically costly (Theorem 3), and, whenever signals satisfy the monotone likelihood ratio property, so are stronger directional motives (Theorem 4).

For the naive motivated reasoner, information is psychologically attractive in expectation. Since the DM evaluates future signals using the motivated prior mean, information is expected to confirm rather than challenge desirable beliefs. As a result, the DM may acquire information even when its direct value is negative. Hence, motivated reasoning does not produce strategic ignorance per se; instead, the latter emerges only when the DM understands how information constrains future motivated beliefs. Importantly, this result is general: from any signal, a naive motivated reasoner expects to gain  $D_{KL}(\tilde{m}||m) \geq 0$  in belief utility, since he anticipates signals generated by his own motivated prior (Theorem 2 in App. B).

Furthermore, Little (2025) shows that a once-motivated reasoner has a Fully Bayesian Equivalent, that is, a Bayesian with prior  $\tilde{f}_0$  who holds identical posteriors after any signal. In the normal-linear case, a Bayesian with prior  $\mathcal{N}(\tilde{\mu}_0, \sigma_0^2)$  has posterior mean  $\lambda s + (1 - \lambda)\tilde{\mu}_0 = \mu_s + \alpha\sigma_s^2 = \tilde{\mu}_s$  and variance  $\sigma_s^2$ . Hence belief updating alone cannot distinguish motivated reasoning from Bayesian updating. However, information acquisition can: the Fully Bayesian Equivalent in Little (2025), which has no belief utility, acquires any signal with  $b \geq 0$ , whereas the sophisticated motivated reasoner in our model requires  $b \geq \frac{1}{2}\alpha^2\lambda\sigma_0^2 > 0$ . At the same time, the naive motivated reasoner acquires whenever the Fully Bayesian Equivalent does, and even some signals with  $b < 0$  that it would reject. Thus, although once-motivated reasoning is indistinguishable from Bayesian updating in how beliefs respond to signals, it can be distinguished from its Fully Bayesian Equivalent through information acquisition when net acquisition benefits are held fixed, and in opposite directions depending on awareness: sophisticated motivated reasoners avoid valuable information, while naive ones seek information of negative value.

### 3.3 Asymmetry-sensitive Motivated Reasoner

Now let us consider a DM who anticipates an asymmetry in information valence. That is, the DM expects future information to be unfavorable on net. This introduces an additional channel through which information acquisition may become costly.

Formally, let  $\delta \geq 0$  denote the anticipated unfavorable shift in the expected signal, measured in the same units as  $s$ . Thus, relative to the benchmark expectation  $\mu \in \{\mu_0, \tilde{\mu}_0\}$ , the asymmetry-sensitive DM evaluates the expected signal as  $\mathbb{E}_s[s] = \mu - \delta$ . This specification captures the case in which the DM anticipates that the information environment may be psychologically unfavorable on net. The parameter  $\delta$  represents a reduced-form measure of anticipated valence asymmetry.

Since  $\mathbb{E}_s[s] = \mu - \delta$ , where  $\mu \in \{\mu_0, \tilde{\mu}_0\}$ , the expected belief utility with the posterior for the sophisticated asymmetry-sensitive motivated reasoner is:

$$\mathbb{E}_s^S[\tilde{B}(f_s)] = \alpha\mu_0 + \frac{1}{2}\alpha^2(1 - \lambda)\sigma_0^2 - \alpha\lambda\delta. \quad (10)$$

At the same time, since  $\tilde{\mu}_0 = \mu_0 + \alpha\sigma_0^2$ , the expected belief utility with the posterior for the naive asymmetry-sensitive motivated reasoner is:

$$\mathbb{E}_s^N[\tilde{B}(f_s)] = \alpha\mu_0 + \frac{1}{2}\alpha^2(1 + \lambda)\sigma_0^2 - \alpha\lambda\delta. \quad (11)$$

The two expressions (10) and (11) differ by  $\alpha^2\lambda\sigma_0^2$ . Thus, sophisticated and naive asymmetry-sensitive motivated reasoners evaluate information differently. The latter assigns a higher expected utility to information since he expects future beliefs to remain directionally favorable. Relative to the baseline sophisticated and naive cases, information sensitivity introduces the additional term  $-\alpha\lambda\delta$ .

**Proposition 5.** *The sophisticated DM (who is aware of his own directional motive) will acquire a signal  $s$  with overall benefit  $b$  if and only if  $b \geq \frac{1}{2}\alpha^2\lambda\sigma_0^2 + \alpha\lambda\delta$ . The naive DM (who is unaware of his own directional motive) will acquire a signal  $s$  with overall benefit  $b$  if and only if  $b \geq \alpha\lambda\delta - \frac{1}{2}\alpha^2\lambda\sigma_0^2$ .*

The comparative statics differ across the two types of motivated reasoners. For the sophisticated asymmetry-sensitive motivated reasoner, the threshold for information acquisition increases with both  $\alpha$  and  $\delta$ . Stronger directional motives make information less attractive since information reduces uncertainty and therefore limits the ability to sustain desirable beliefs. Likewise, a larger anticipated unfavorable shift increases the psychological cost of learning. Consequently, sophisticated motivated reasoners become less willing to acquire information as either  $\alpha$  or  $\delta$  increases.

For the naive asymmetry-sensitive motivated reasoner, the effect of  $\delta$  is identical: a larger  $\delta$  increases the threshold for information acquisition since the anticipated difference between favorable and unfavorable information becomes more important. The effect of  $\alpha$ , however, is non-monotone: the threshold increases in  $\alpha$  whenever  $\delta > \alpha\sigma_0^2$ , and decreases in  $\alpha$  whenever  $\delta < \alpha\sigma_0^2$ .

Intuitively, stronger directional motives increase the psychological value of maintaining desirable beliefs, but they also increase the extent to which future motivated beliefs benefit the DM. Figure 2 shows that increasing  $\delta$  shifts both acquisition thresholds upward. The distance between the two thresholds remains constant, so information sensitivity does not eliminate the naive–sophisticated gap; instead, it makes information acquisition harder for both types while preserving the strategic-ignorance region in which only the naive type learns.

Overall, asymmetry sensitivity makes information acquisition less attractive because the DM anticipates that learning may expose him to psychologically unfavorable information. For sophisticated motivated reasoners, stronger motives and greater sensitivity to unfavorable information both reduce the willingness to learn. For naive motivated reasoners, sensitivity to unfavorable information also reduces information acquisition, but stronger directional motives have two opposing effects: they raise the value of preserving desirable beliefs while also making the DM more optimistic about what future information will support.

## 4 Discussion

Here, we provide a brief discussion of several aspects of modeling motivated reasoning, as well as the experimental settings for testing the model.

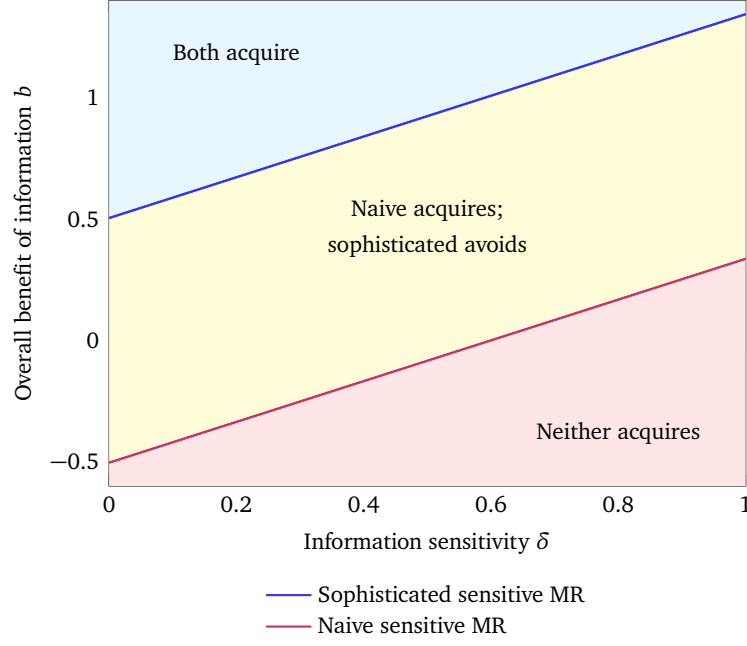


Figure 2: Information sensitivity thresholds in  $(\delta, b)$ -space.

#### 4.1 Testable Hypotheses and Related Literature

In line with the insights provided by the formal model, we produce a range of hypotheses that are directly informed by the model's structure and that can be tested empirically (e.g., through experiment).

First, following Proposition 3, the threshold for information acquisition in the baseline belief-acquisition model increases with the precision of the signal. A more precise signal provides less flexibility for motivated belief formation, making it less desirable for a sophisticated (self-aware) motivated reasoner to acquire the signal. Consequently, the model predicts that individuals will be less willing to acquire highly informative signals. Importantly, this prediction pertains to the baseline model in which information is valued through its effect on beliefs. In the action-based extension, this effect disappears because both the objective decision benefit and the psychological cost of information scale proportionally with the reduction in posterior variance.

**Hypothesis 1** (Signal precision). *In the baseline belief-acquisition model, sophisticated individuals (who are aware of their own self-serving biases) will be less likely to acquire information as the precision of the signal ( $\lambda$ ) increases.*<sup>9</sup>

Related evidence from political science shows that the influence of informational content depends on the political environment. In two survey experiments on offshore drilling and immigration, Druckman et al. (2013) find that substantive arguments exert significant influence on opinions in non-polarized environments, whereas elite polarization increases reliance on party endorsements and reduces the influence of information. Thus, the study demonstrates that the

<sup>9</sup>By Theorem 3 in Appendix B, **H(1)** holds for any pair of Blackwell-ranked signals (not only for normal signals of different precision).

informational content of a signal and the political context in which it is received jointly shape political opinion formation, but it does not manipulate signal precision or citizens' willingness to acquire the signal.

From an economics perspective, **H(1)** is broadly consistent with recent experimental studies suggesting that motivated distortions become more pronounced when signals are less informative or more ambiguous. For instance, [Engelmann et al. \(2024\)](#) test how subjects undertake pattern recognition tasks (where certain patterns could lead to an electric shock or financial loss). The tendency to engage with wishful thinking—i.e., as the authors define it, self-deception motivated by anticipatory utility concerns (e.g., by a desire to feel better about the future)<sup>10</sup>—becomes stronger when signals are less precise. More specifically, [Engelmann et al. \(2024\)](#) document that wishful thinking becomes more evident when participants face more ambiguous or challenging patterns, since these produce weaker signals that, by construction, are easier to interpret as more favorable outcomes to a subject<sup>11</sup>.

Second, Corollary 2 (in Appendix A) shows that sophisticated motivated DMs will acquire a signal  $s$  if and only if  $\ell \geq \frac{1}{2}\alpha^2$ . Since the threshold for information acquisition increases with the strength of the directional motives ( $\alpha$ ), the model predicts that more motivated individuals will be less inclined to seek information.

**Hypothesis 2** (Directional motives). *Sophisticated individuals (who are aware of their own self-serving biases) with higher directional motives (higher  $\alpha$ ) will be less likely to acquire information compared to those with lower directional motives.*<sup>12</sup>

Closely related evidence in political science appears in experimental research on partisan selective exposure. Among the most relevant works, [Iyengar and Hahn \(2009\)](#) experimentally vary the news source attached to otherwise identical headlines and show that news demand responds systematically to partisan affinity. Namely, Republicans and conservatives prefer reports attributed to Fox News and avoid CNN and NPR, whereas Democrats and liberals avoid Fox News. Importantly, this tendency to select information on the basis of anticipated partisan agreement is stronger among more politically engaged partisans. Relatedly, [Taber and Lodge \(2006\)](#) find that motivated processing is strongest among individuals with stronger prior attitudes, who selectively accept congenial evidence and counterargue uncongenial evidence, consistent with the prediction that stronger directional motives intensify resistance to threatening information. Using a dynamic information-board experiment that allowed subjects to actively search for information during a simulated presidential campaign, [Redlawsk \(2002\)](#) likewise finds that information search is systematically directed by prior affect. Subjects disproportionately search for information about candidates they already like while neglecting candidates they dislike. Moreover, affectively incongruent information requires greater processing time, and among motivated

<sup>10</sup>Namely, [Engelmann et al. \(2024\)](#) define wishful thinking as “self-deception that is driven by a desire to feel better about the future” (p. 927), or “self-deception motivated by anticipatory utility concerns” (p. 929).

<sup>11</sup>The statistical analysis of the difficulty coefficient across different patterns indicates that subjects are less accurate with harder patterns. The variability of interaction coefficients—i.e., Shock/Loss pattern  $\times$  HAB or Shock/Loss pattern  $\times$  DP (see even-numbered columns in Table 2 (p. 940) in [Engelmann et al., 2024](#))—demonstrates that the impact of loss or shock patterns intensifies as difficulty increases.

<sup>12</sup>Theorem 4 from Appendix B shows that this prediction is not an artifact of the normal-linear specification.

reasoners greater incongruence can strengthen prior candidate evaluations. These results are consistent with **H(2)** in that politically relevant directional considerations shape the decision of which information to acquire.

From an experimental economics literature perspective, **Drobner and Goerg (2024)** introduce a mechanism that enables researchers to manipulate the extent of perceived ego-relevance, which, in turn, affects the subjects' ex ante degree of motivation or degree of directional motives.<sup>13</sup> Eventually, they find that subjects in the High-Ego treatment exhibited overconfident prior beliefs and ended with more optimistic posterior beliefs compared to those in the Low-Ego treatment.<sup>14</sup>

Third, for  $\alpha > 0$  and  $\lambda > 0$ , Proposition 3 and Proposition 4 imply that sophisticated motivated reasoners face a strictly positive threshold for acquiring information, whereas naive motivated reasoners may acquire the signal even when its direct benefit is slightly negative. Therefore, sophisticated motivated reasoners will engage in more information avoidance compared to naive motivated reasoners.

**Hypothesis 3** (Sophisticated vs. naive motivated reasoner). *Sophisticated individuals (who are aware of their own self-serving biases) are less likely to acquire information than naive (i.e., unaware) ones.*

To the best of our knowledge, **H(3)** has not been directly tested in the existing political-science or economics literature. Related studies nevertheless show that greater political engagement or cognitive sophistication does not necessarily eliminate directional information processing. **Iyengar and Hahn (2009)**, for example, find that partisan selective exposure is stronger among more politically engaged partisans, while **Kahan (2013)** finds that individuals scoring higher in cognitive reflection can display greater ideological polarization in their assessment of politically contested evidence. These findings are consistent with the broader possibility that sophistication need not attenuate motivated cognition, but neither study identifies the mechanism proposed here: awareness of one's own directional motive and anticipation of its consequences for future beliefs. Distinguishing such self-awareness from political knowledge, engagement, or cognitive ability is therefore essential for a direct test of **H(3)**.

Fourth, Corollary 4 shows that the threshold for acquiring information increases with the DM's sensitivity to the difference between favorable and unfavorable signals ( $\delta$ ). Therefore, the model predicts that DMs who react more strongly to the difference between good and bad news will be less likely to seek information.

**Hypothesis 4** (Good vs. bad news asymmetry). *Individuals—both sophisticated and naive—who anticipate a larger unfavorable shift in information (higher  $\delta$ ) will be less likely to acquire information compared to those who anticipate a smaller shift.*

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<sup>13</sup>To do so **Drobner and Goerg (2024)** provide participants with different articles summarizing scientific evidence regarding the importance of IQ tests: half of the sample read about the strong correlation between IQ and ego-relevant outcomes like income and health (the High-Ego treatment) and the other half read about the evidence against IQ as a measure of intelligence (the Low-Ego treatment).

<sup>14</sup>They also updated their beliefs more optimistically, showed a higher tendency to make incorrect or no updates when faced with negative signals, and rationalized negative feedback by diminishing the importance of the IQ test and downplaying their effort, whereas in the Low-Ego treatment, updates were consistent regardless of signal valence.

Particularly direct political-behavior evidence comes from [Heseltine et al. \(2025\)](#), who study nine months of online browsing behavior comprising 27,648,770 visits by 2,462 American adults. They show that partisan news consumption changes dynamically with the valence of political news. At the aggregate level, Democrats do not significantly change their overall news consumption when daily coverage becomes more negative toward their in-party, but they reduce their consumption of hard news; Republicans instead slightly increase news consumption on such days, primarily by shifting toward ideologically aligned partisan outlets. At the browsing-session level, encountering negative coverage of either party reduces subsequent browsing length for both Democrats and Republicans, with the reduction strongest following negative coverage of the respondent’s own party.

From an economics point of view, the major part of the literature on motivated reasoning shows that the good vs. bad news asymmetry is present, which gives rather strong evidence for **H(4)**. That is, subjects update (significantly) much more under favorable signals than under unfavorable ones as long as these signals are not orthogonal to the DM’s identity. The asymmetry of updating is a rather robust finding which is captured by many designs and specifications (e.g., [Zimmermann, 2020](#); [Drobner, 2022](#); [Thaler, 2024a](#)). Since intellectual identity is often salient for Western subjects, one widespread context in which asymmetric belief updating is tested is performance on IQ tests (e.g., Raven matrices). The presence of identity relevance seems to be the major factor of motivated reasoning. While it is not yet perfectly clear<sup>15</sup>, the valence (positive vs. negative) of news in the context of financial decision-making—regardless of the subject’s identity—does not appear to cause motivated reasoning (e.g., [Barron, 2021](#)). Similarly, the valence of news—whether independent of or relevant to their functional values—in the context of message trust does not lead to motivated reasoning (e.g., [Thaler, 2024b](#)).

Fifth, in the asymmetry-sensitive action extension (Corollary 4 in Appendix A), the threshold for information acquisition depends on prior uncertainty through the term  $\alpha\delta\sigma_0^{-2}$ . Since this term decreases as  $\sigma_0^2$  increases, greater prior uncertainty lowers the threshold for acquiring information. Therefore, the DM with greater dispersion of his prior beliefs (i.e., higher variance  $\sigma_0^2$ ) faces a lower threshold for acquiring information.

**Hypothesis 5** (Prior confidence). *In the asymmetry-sensitive action extension, for  $\alpha\delta > 0$ , individuals—both sophisticated and naive—with stronger prior beliefs (lower  $\sigma_0^2$ ) are less likely to seek new information compared to those with more uncertain prior beliefs (higher  $\sigma_0^2$ ).*

For instance, [Kahan et al. \(2012\)](#) show that citizens with greater scientific literacy and numeracy become more rather than less polarized on climate-change risks, demonstrating that greater cognitive sophistication does not necessarily mitigate politically motivated disagreement. Another related political-science result concerns the endogenous formation of confidence rather than the causal effect of prior confidence on information acquisition. In two survey experiments, [Druckman et al. \(2013\)](#) find that polarized partisan environments increase citizens’ confidence

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<sup>15</sup>While the financial domain does not inherently evoke the relevance to subjects that triggers motivated reasoning, an interesting direction for research—both fundamentally significant and methodologically sound—would be to investigate the specific mechanisms through which a financial context may induce motivated reasoning.

in policy opinions even as those opinions become less responsive to substantive arguments. The authors further argue that such increased confidence can make individuals less likely to consider alternative positions. While these findings are related to **H(5)**, they do not directly test the role of prior confidence in information acquisition.

From an economics perspective, while dealing with a novel approach to identifying motivated reasoning, **Thaler (2024a)** addresses the issue of the (over-)precision of prior beliefs. First, he shows that the results on overprecision indicate that subjects often overestimate the accuracy of their beliefs, particularly on politicized topics, with confidence intervals containing the correct answer less than 50 percent of the time. This overprecision is more pronounced in partisans than in moderates, supporting the idea that individuals with strong prior beliefs (lower  $\sigma_0^2$ ) might be overconfident in their assessments. Consequently, overprecision may reduce their likelihood of seeking new information, as they perceive their current beliefs to be sufficiently accurate. Furthermore, **Thaler (2024a)** notes that overprecise individuals are more likely to trust error-reinforcing Fake News and are less likely to trust True News compared to underprecise individuals, suggesting that motivated beliefs can shape how individuals process and assess information. This behavior aligns with **H(5)**.

Sixth, following Propositions 3, 4, and 5, higher exogenous benefits from information acquisition increase the likelihood that a DM chooses to acquire information.

**Hypothesis 6** (Value of higher incentives). *For  $\gamma > 0$ , individuals—both sophisticated and naive—will be more likely to acquire information when the objective value of information,  $b^{obj}$ , is higher.*

Political-science experiments show that increasing the value of accuracy can substantially change behavior in politically charged informational environments, although existing studies typically examine reported beliefs rather than information acquisition itself. In two experiments, **Bullock et al. (2015)** randomly provide monetary incentives for accurate responses to partisan factual questions and find that these incentives substantially reduce partisan divergence. Across questions displaying an unincentivized partisan gap, incentives for correct answers reduce the gap by approximately 55–60 percent, while a treatment rewarding both correct answers and admissions of “don’t know” reduces the gap by approximately 80 percent. Importantly, the distinction between incentivizing accurate responses and incentivizing information acquisition is supported by **Peterson and Iyengar (2021)**. In experiments that both incentivize correct answers and allow respondents to search for information before answering political factual questions, they find that accuracy incentives reduce some partisan divergence in reported factual beliefs but have no effect on partisan differences in information search. A complementary result comes from **Bolsen et al. (2014)**, who experimentally induce an accuracy motivation by informing respondents that they will need to justify their policy opinions. In their survey experiment on energy policy, partisan motivated reasoning appears under in-party and out-party endorsements but disappears when respondents are induced to pursue an accuracy goal. Taken together, these results reinforce the distinction between directional and accuracy motives while also highlighting that accuracy-motivated responding and information acquisition are distinct behavioral margins.

From an economics point of view, there is also some evidence supporting **H(6)**, although the

results remain somewhat mixed. For example, [Engelmann et al. \(2024\)](#) aim to find the effect of stakes on accuracy. In the first two experiments (experiments 1 and 2), [Engelmann et al. \(2024\)](#) use patterns that are either tilted to the right or left (commonly referred to as Gabor patterns in the paper). Since these patterns may be too specific to capture a general behavioral phenomenon, the authors introduce two additional experiments. In these experiments, they replace the initial visual task of identifying a single right-or-left tilted pattern in a picture (a single Gabor flash) with a series of eight such pictures (eight Gabor flashes in experiment 3) and a square consisting of red and blue dots that exhibits another type of pattern (colored dots in experiment 4).<sup>16</sup> Based on these latter types of visual tasks, [Engelmann et al. \(2024\)](#) explicitly examine the effect of stake sizes (namely: € 1 vs. € 20 in experiment 1, £0.1 vs. £10 in experiment 2, £0.05 vs. £10 in experiment 3, and £0.05 vs. £10 in experiment 4) on the accuracy of pattern recognition. In all four experiments, for both OLS specifications—with and without interaction terms—the effect of the high accuracy bonus is not significant, except in one case: only the regression specification without the interaction term shows the significant effect of higher stakes on accuracy.

Seventh, suppose that the objective value of information exceeds its psychological value,  $b^{\text{obj}} > b^{\text{psy}}$ . Then increasing the weight  $\gamma$  on the objective value of information raises the overall benefit of acquisition,  $b = \gamma b^{\text{obj}} + (1 - \gamma) b^{\text{psy}}$ , and therefore increases the likelihood that the DM acquires information.

**Hypothesis 7** (Trade-off between objective and subjective value of information). *When  $b^{\text{obj}} > b^{\text{psy}}$ , individuals—both sophisticated and naive—who assign greater weight to the objective value of information (higher  $\gamma$ ) will be more likely to acquire information.*

The discussion of this hypothesis is closely related to that of **H(6)**; however, additional insights can be drawn in connection with **H(7)**. Political-science evidence also demonstrates directly that information choice reflects competing components of the value of news. In a conjoint survey experiment, [Mummolo \(2016\)](#) repeatedly presents partisans with pairs of news items while independently randomizing both the partisan reputation of the source and whether the topic is relevant to the respondent. Politically unfriendly sources reduce the probability that an item is selected, but topic relevance often exerts a substantially larger positive effect: across five of six respondent subgroups, relevant topics increase selection by approximately 8 to 30 percentage points, frequently outweighing the penalty associated with a politically dissonant source. This provides a close empirical analogue to **H(7)**, while not identifying the model's parameter  $\gamma$  directly.

From an economics perspective, one rather popular method is based on an incentive-compatible mechanism introduced by [Becker et al. \(1964\)](#). For instance, [Engelmann et al. \(2024\)](#) use it to estimate the individual value of having distorted beliefs; or, as another somewhat relevant example, [Guan et al. \(2023\)](#), interested in measuring preferences for informativeness, derive strict cardinal preferences by assessing subjects' willingness to pay for access to the information structure that impacts their forthcoming decisions.

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<sup>16</sup>Examples for all three types of visual tasks can be found on p. 933 in [Engelmann et al. \(2024\)](#).

## 4.2 Modelling Motivated Reasoning

So far, neither the political science literature nor economics literature appears to have reached a consensus on which approach can be considered the benchmark. While alternative formalizations are also feasible (a relevant discussion can be found in, e.g., [Bracha and Brown, 2012](#); [Mayraz, 2019](#); [Little, 2025](#)), the modeling approach to motivated reasoning employed in this paper offers several advantages.

Building on [Little \(2025\)](#), we shift attention from how motivated individuals process information after receiving it to whether they choose to acquire information in the first place. This distinction connects the model to a political-behavior literature showing that partisan motives operate not only through the interpretation of information after exposure, but also through endogenous information search, selective exposure, and avoidance ([Redlawsk, 2002](#); [Iyengar and Hahn, 2009](#); [Arceneaux et al., 2012](#); [Mummolo, 2016](#); [Peterson and Iyengar, 2021](#); [Peterson and Kagalwala, 2021](#); [Heseltine et al., 2025](#)).<sup>17</sup>

The central distinction in our model is between motivated reasoners who anticipate the consequences of their directional motives for future belief formation and motivated reasoners who do not. This distinction drives the paper’s main result: motivated reasoning alone does not yield strategic ignorance; rather, strategic ignorance emerges when individuals recognize that future information may constrain psychologically valuable beliefs.

### 4.2.1 Advantages

First, our approach identifies a novel mechanism linking political sophistication to information avoidance. Existing theories typically explain motivated reasoning through biased processing of information. By contrast, our model shows that strategic ignorance can emerge before information is even observed. This provides a formal explanation for why politically sophisticated citizens may sometimes be less willing to engage with belief-threatening information.

Second, our approach, based on the framework developed by [Little \(2025\)](#), encompasses several behavioral regularities documented in recent literature, both in political psychology and behavioral economics. While the comprehensive version of [Little \(2025\)](#) (known as *twice-motivated reasoning*) addresses various empirical and theoretical findings (e.g., disputes regarding the quality of information linked to prior beliefs, variations in disagreement influenced by directional or accuracy primes, and asymmetric updating in response to “good” versus “bad” news), a simpler model (referred to as *once-motivated reasoning*) is observationally equivalent to Bayesian updating in how beliefs respond to signals. Our results show that this equivalence does not extend to information acquisition: a sophisticated once-motivated reasoner avoids information that its Fully Bayesian Equivalent in [Little \(2025\)](#) would acquire. The simplest formulation of motivated reasoning can thus be distinguished from its Fully Bayesian Equivalent through the demand for

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<sup>17</sup>At the same time, this literature does not imply universal avoidance of cross-cutting information: selective exposure is conditional on features of the information environment and the value of the available content, with topic relevance sometimes outweighing partisan source preferences ([Mummolo, 2016](#)). Our approach is consistent with this conditional view because acquisition depends on the psychological cost of information relative to its other benefits.

information, provided that net acquisition benefits are held fixed.

Third, our approach enables us to derive several testable predictions. Some of these predictions have already been examined in the literature and support the hypotheses outlined in the model presented in this paper, demonstrating the relevance of our theory in bridging behavioral regularities with a formal theoretical framework. Other predictions are novel contributions to the literature and offer potential avenues for future research.

#### 4.2.2 Assumptions

As assumptions constitute an important part of any formal modeling, we discuss the most crucial ones here.

***Unawareness vs. awareness of own biases.*** One key assumption is the possibility of distinguishing DMs who are self-aware of self-serving biases, whom we call sophisticated ones, from those who are unaware, whom we call naive. This distinction is crucial for understanding how people process information, especially in political (or other identity-relevant) contexts. Self-aware motivated reasoners recognize that their reasoning may be influenced by their biases or desires; however, this awareness does not guarantee the ability to overcome these biases. They may still engage in motivated reasoning but do so with an understanding of the factors influencing them. In contrast, unaware reasoners remain oblivious to the influence of their biases, genuinely believing they are evaluating evidence objectively, even when their reasoning is shaped by self-serving motives.

***Sensitivity to asymmetry of information valence.*** The case where individuals place greater weight on “good news” signals is well-documented in both political-psychology and behavioral-sciences literatures. A notable example, increasingly studied in experimental economics, involves subjects who are asked to complete a series of cognitive tasks and subsequently rank themselves among other participants. This is often relevant to their identity, such as considering themselves intellectuals (e.g., Zimmermann, 2020; Drobner, 2022). In contrast, the situation where individuals assign more weight to “bad news” signals is less represented in both political science and economics literatures. This scenario often arises when a subject’s identity is tied to negative feedback.<sup>18</sup>

***Objective vs. subjective value of information.*** A substantial body of literature explicitly documents that subjects care not only about the objective value ( $b^{\text{obj}}$ ) of information but also about the potential psychological effects ( $b^{\text{psy}}$ ) of encountering and incorporating it (e.g., Golman et al., 2017; Benjamin, 2019; Sharot and Sunstein, 2020; Stanovich, 2021; Kelly and Sharot, 2021; Molinaro et al., 2023; Vu et al., 2023). This holds true in the context of motivated reasoning as well (e.g., Dimant et al., 2024; Chen and Heese, 2026).

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<sup>18</sup>For instance, consider a peasant community where a strong social norm dictates staying connected to the land and discourages leaving it for any reason, such as pursuing education. In this context, receiving objectively negative news, such as failing a university entrance exam, is likely to be weighted more heavily by a peasant who does not want to leave his land, as it aligns with his desire to remain connected to his community.

### 4.3 Further Notes on the Experiment Design

**Prior and posterior belief distributions.** Using the experimental framework for belief-updating experiments, we can easily collect the whole distribution of both prior and posterior beliefs. For example, in the context of motivated reasoning Zimmermann (2020) collects the whole prior and posterior distribution of beliefs about subjects' position in the ranking based on their performance in the IQ test. Using similar procedures would allow a researcher to recover both the prior mean  $\mu_0$  and the prior variance  $\sigma_0^2$ . An alternative, though more demanding, approach is to elicit entire belief distributions graphically, as in Gonzalez-Fernandez (2024). This allows subjects to allocate probability mass across possible states directly, thereby recovering the shape and dispersion of the belief distribution rather than relying only on discrete probability reports.

**Eliciting subjects' (un)awareness of own directional motives.** One straightforward way to manipulate awareness is to vary the level of information provided about the issue of motivated reasoning, or about different (belief) biases in general. In this context, using a conventional design capable of eliciting motivated reasoning and incorporating two treatments can serve to test H(3). While details of the experimental design will need some refinement, the general idea is relatively simple: in a treatment (compared to the baseline with no information on that matter), subjects are provided with a brief introduction to cognitive biases that humans are generally susceptible to, the consequences of relying on these biases in decision-making, and general strategies to mitigate them<sup>19</sup>. Naturally, embedding initial (and subsequent) measures before (and after) introducing information on biases may be essential. Additionally, beyond directly structuring the treatments, measures of reaction or response time—which are gaining popularity—can be employed as a useful metric.

**Eliciting subjects' psychological value of information.** While operationalizing the benefit  $b$  of information acquisition within a Bayesian belief updating framework can be challenging—since it must capture a clear and salient advantage for the DM—this can be accomplished in the conventional experimental economics framework through monetary rewards. To assess the effect of the objective value of information ( $b^{\text{obj}}$ ), a straightforward approach is to use two treatments with high and low payoffs. In a recent study, Enke et al. (2023) investigate the impact of incentives on four well-documented biases—anchoring, base-rate neglect, failure of contingent thinking, and intuitive reasoning—and find that while very high incentives result in only minor improvements in performance, they do increase response times by 40 percent. To capture the subjective value of information ( $b^{\text{psy}}$ ), an experimenter can design the incentives to reflect subjects' willingness to pay. For instance, Guan et al. (2023) examine subjects' demand for information—shaped by the *informativeness* of information structures—in an abstract laboratory experiment context.

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<sup>19</sup>Depending on the specific research question, researchers may find it more useful and precise to separate the testing of each element. For example, disentangling the introduction of biases and explanation of their consequences from explanations of optimal solutions and ways to avoid these biases within individual treatments could comprise two separate treatments.

## 5 Conclusion

In this paper, building on the framework of [Little \(2025\)](#), we develop a formal theory to answer the question of why awareness of one's own motivated reasoning—viewed as part of cognitive sophistication—can reduce information acquisition. The central distinction in our framework is between sophisticated motivated reasoners, who are aware of their own directional motives, and naive motivated reasoners, who are unaware of them.

Our analysis yields several results. First, we demonstrate that motivated reasoning alone does not produce strategic ignorance. Naive motivated reasoners remain willing to acquire information since they fail to anticipate how self-serving biases will shape their future beliefs. By contrast, sophisticated motivated reasoners recognize that information reduces uncertainty and thereby narrows the opportunity for maintaining directionally desirable beliefs. Second, we show that more informative signals may become less attractive to those motivated reasoners who are self-aware. Third, we show that sensitivity to the asymmetry between favorable and unfavorable information further strengthens incentives to avoid learning. Finally, when beliefs guide actions, awareness creates a decision-payoff tradeoff: sophisticated motivated reasoners may forgo information that would objectively improve their decisions, whereas naive motivated reasoners always acquire it in the action benchmark without valence asymmetry.

These findings have important implications for political behavior and democratic theory. A general expectation in empirical research as well as a common assumption in formal papers is that greater sophistication improves citizens' ability to process political information and hold political elites accountable. Our results emphasize the fragility of this assumption: simply increasing the availability or quality of political information may be insufficient for improving democratic accountability. Policies designed to improve democratic decision-making may therefore need to address not only the supply of information but also the psychological incentives governing whether citizens choose to acquire it.

Several directions for future research appear promising. First, the model can be extended to alternative belief distributions, richer information structures, and environments with multiple competing signals. Second, the model produces a number of testable predictions, at least one of which appears to be unexplored in the empirical literature. More broadly, our analysis suggests directing greater attention toward how factual information should be communicated when identity-relevant motives create incentives to avoid it.

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# Appendix

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## A Extension: Mapping Beliefs to Actions

Suppose now that the DM can anticipate making better decisions with more information. Consider the DM taking an action  $a$  with a separate decision utility:

$$u(a, \omega) = -\ell(a - \omega)^2, \quad (12)$$

where  $a$  is the action chosen by the DM,  $\omega$  is the true state of the world, and  $\ell > 0$  represents the loss parameter, capturing the cost of making an incorrect decision. This is a standard quadratic-loss specification in which deviations of the action from the true state are penalized with weight  $\ell$  (see, e.g., [Savage, 1972](#); [Raiffa and Schlaifer, 2000](#); [DeGroot, 2005/1970](#)).

Conditional on the signal  $s$ , the objective posterior is normal with mean  $\mu_s$  and variance  $\sigma_s^2$ , so the expected decision utility of action  $a$  is

$$\mathbb{E}_\omega[-\ell(a - \omega)^2 | s] = -\ell(a - \mu_s)^2 - \ell\sigma_s^2, \quad (13)$$

since  $\mathbb{E}_\omega[(a - \omega)^2 | s] = (a - \mu_s)^2 + \mathbb{V}[\omega | s]$ .

When taking an action equal to the objective posterior mean, the expected decision utility becomes equal to  $-\ell\sigma_s^2$ . In what follows, the decision value of information is evaluated using the objective posterior mean. If instead the DM chooses the motivated posterior mean as the action while losses are evaluated relative to the true state, an additional distortion cost appears. Specifically, if  $a = \tilde{\mu}_s = \mu_s + \alpha\sigma_s^2$ , then  $\mathbb{E}_\omega[u(a, \omega) | s] = -\ell(\alpha^2\sigma_s^4 + \sigma_s^2)$ , since  $\tilde{\sigma}_s^2 = \sigma_s^2$ . Thus, choosing the motivated posterior mean as an action introduces an additional distortion cost equal to  $\ell\alpha^2\sigma_s^4$ , since  $\mathbb{E}_\omega[u(\mu_s, \omega) | s] = -\ell\sigma_s^2$  and  $\mathbb{E}_\omega[u(\tilde{\mu}_s, \omega) | s] = -\ell(\alpha^2\sigma_s^4 + \sigma_s^2)$ .

The objective value of information under quadratic loss is the reduction in expected loss produced by the signal. Since the posterior variance is  $\sigma_s^2 = (1 - \lambda)\sigma_0^2$ , the improvement in expected decision utility from observing the signal is  $b = \ell(\sigma_0^2 - \sigma_s^2) = \ell\lambda\sigma_0^2$ . Thus, throughout this extension, we interpret the objective benefit of information as the reduction in expected decision loss:  $b = \ell\lambda\sigma_0^2$ .

### A.1 Sophisticated Motivated Reasoner

First, let us (re)consider the DM who is self-aware of his own directional motive.

**Corollary 2.** *The sophisticated DM (who is self-aware of his own directional motive) will acquire a signal  $s$  with the loss parameter  $\ell$  if and only if  $\ell \geq \frac{1}{2}\alpha^2$ .*

The main feature of comparative statics in Corollary 2 is that the threshold value of the loss parameter increases with  $\alpha$ , which implies that more motivated subjects are less likely to seek information. This reflects the idea that stronger directional motives increase the psychological cost of information since informative signals reduce the scope for maintaining desirable beliefs. Thus, DMs will acquire information when the benefit of making better decisions outweighs the cost associated with the potential for less pleasant beliefs.

## A.2 Naive Motivated Reasoner

Second, let us turn to the naive DM who is unaware of his own directional motive.

**Corollary 3.** *The naive DM (who is unaware of his own directional motives) will acquire a signal  $s$  with the loss parameter  $\ell$  if and only if  $\ell \geq -\frac{1}{2}\alpha^2$ .*

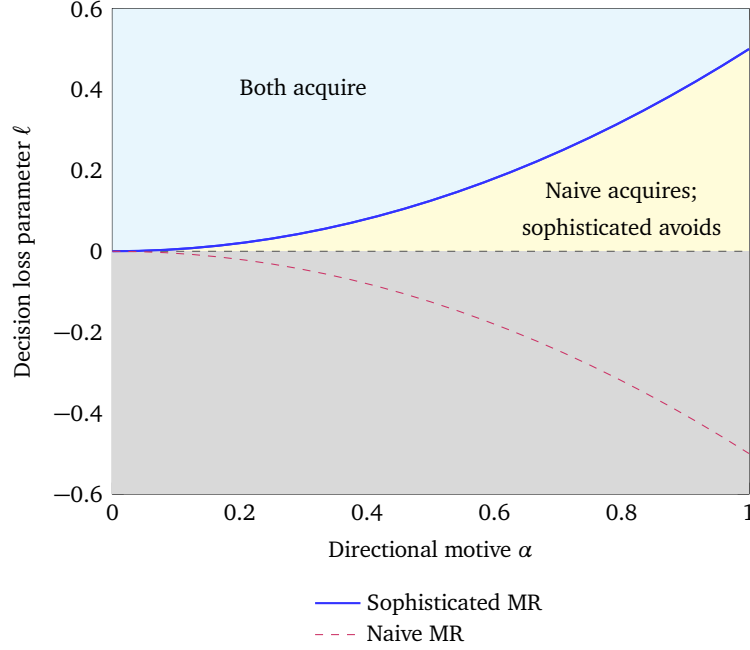


Figure 3: Information acquisition thresholds in  $(\alpha, \ell)$ -space when beliefs map into actions. The grey area depicts the infeasible region, since  $\ell > 0$  by definition in (12).

Although the acquisition threshold formally depends on the strength of directional motives, the threshold is nonpositive and is strictly negative whenever  $\alpha > 0$ . Since the loss parameter satisfies  $\ell > 0$ , the acquisition condition is therefore always satisfied. Hence, naive motivated reasoners always acquire information. Intuitively, naive motivated reasoners acquire information since its positive objective decision benefit is reinforced by their positive expected change in belief utility. Figure 3 illustrates the decision-payoff version of the awareness paradox. Even when information objectively improves subsequent decisions, the naive motivated reasoner acquires it, whereas the sophisticated motivated reasoner may avoid it since he anticipates the psychological cost of holding less desirable beliefs.

## A.3 Asymmetry-sensitive Motivated Reasoner

Finally, let us consider the DM who is asymmetry-sensitive.

**Corollary 4.** *The sophisticated DM (who is aware of his own directional motive) will acquire a signal  $s$  with the loss parameter  $\ell$  if and only if  $\ell \geq \frac{1}{2}\alpha^2 + \alpha\delta\sigma_0^{-2}$ . The naive DM (who is unaware of his own directional motive) will acquire a signal  $s$  with the loss parameter  $\ell$  if and only if  $\ell \geq \alpha\delta\sigma_0^{-2} - \frac{1}{2}\alpha^2$ .*

The first feature of the comparative statics is that anticipated valence asymmetry introduces an additional psychological cost of information. Since  $\mathbb{E}_s^S[s] = \mu_0 - \delta$  and  $\mathbb{E}_s^N[s] = \tilde{\mu}_0 - \delta$ , the anticipated unfavorable shift contributes the additional term  $\alpha\lambda\delta$  to the acquisition threshold, over and above the effect of reduced uncertainty. This follows from the law of iterated expectations: before observing the signal, the expected posterior mean coincides with the prior mean. Information therefore does not change expected beliefs on average, but it does reduce uncertainty, thereby limiting the scope for maintaining desirable motivated beliefs.

The second feature of comparative statics in Corollary 4 concerns the role of directional motives. For the sophisticated asymmetry-sensitive motivated reasoner, the threshold for information acquisition increases monotonically with  $\alpha$ . Thus, stronger directional motives make information acquisition less attractive.

For the unaware asymmetry-sensitive motivated reasoner, however, the effect of directional motives is non-monotonic: the information-acquisition threshold increases with  $\alpha$  whenever  $\alpha < \delta\sigma_0^{-2}$ , and decreases with  $\alpha$  whenever  $\alpha > \delta\sigma_0^{-2}$ . Intuitively, stronger directional motives initially increase the value of favorable beliefs and therefore make information less attractive. However, once directional motives become sufficiently strong, the naive DM becomes more optimistic about the beliefs that future information will support, making information acquisition more attractive. The threshold also increases with the anticipated unfavorable shift  $\delta$ , which means motivated reasoners who are more sensitive to the difference between good and bad news are less likely to choose to receive information.

Finally, for  $\alpha\delta > 0$ , the threshold value of  $\ell$  decreases in  $\sigma_0^2$ : greater prior uncertainty therefore makes the acquisition condition easier to satisfy. This means that as the DM becomes less certain about their prior beliefs (higher uncertainty), they are more willing to acquire new information because the required loss parameter  $\ell$  decreases. In other words, the more confident a DM is in his prior beliefs (lower  $\sigma_0^2$ ), the less he feels the need for new information.

## B General Results

Throughout this appendix, let  $f_0$  be any prior density on  $\Omega$ , let  $v : \Omega \rightarrow \mathbb{R}$  be any directional motive with  $Z_0 \equiv \int f_0(\omega) \exp(v(\omega)) d\omega < \infty$  and  $\int f_0(\omega) \exp(v(\omega)) |v(\omega)| d\omega < \infty$ , and let  $s$  be any signal with conditional density  $p(s | \omega)$ , posterior  $f_s(\omega) = p(s | \omega) f_0(\omega) / m(s)$ , and objective predictive density  $m(s) = \int p(s | \omega) f_0(\omega) d\omega$ . By Proposition 1, the motivated prior is  $\tilde{f}_0(\omega) = f_0(\omega) \exp(v(\omega)) / Z_0$ ; we denote the corresponding *motivated predictive density* by  $\tilde{m}(s) = \int p(s | \omega) \tilde{f}_0(\omega) d\omega$ .

Consistent with Section 3, the sophisticated DM evaluates signals under  $m$ , whereas the naive DM evaluates them under  $\tilde{m}$ . This generalizes Section 3.2: in the normal-linear case,  $\tilde{B}(f_s)$  is linear in  $s$ , so only  $\mathbb{E}_{\tilde{m}}[s] = \tilde{\mu}_0$  matters (see Corollary 5). Since  $\exp(v) > 0$ ,  $m$  and  $\tilde{m}$  are mutually absolutely continuous, so both Kullback–Leibler divergences below are well defined in  $[0, \infty]$ . All expectations below are well defined, since  $(\log x)^+ \leq x$  and  $\mathbb{E}_m[\tilde{m}/m] = \mathbb{E}_{\tilde{m}}[m/\tilde{m}] = 1$  bound the positive parts of the relevant log-ratios.

**Theorem 1** (Sophisticated motivated reasoners weakly dislike information). *For a sophisticated DM, who evaluates signals under the objective prior, the psychological cost of information*

$$C \equiv \tilde{B}(f_0) - \mathbb{E}_m[\tilde{B}(f_s)]$$

*satisfies  $C \geq 0$ , with equality if and only if  $\int f_s(\omega) \exp(v(\omega)) d\omega$  is constant for  $m$ -almost every  $s$ . Consequently, the sophisticated DM acquires the signal if and only if  $b \geq C$ , and never acquires a signal with  $b < 0$ .*

**Theorem 2** (Value of information as predictive divergence). *The expected change in belief utility from acquiring the signal is*

$$\begin{aligned} (\text{sophisticated DM}) \quad & \mathbb{E}_m[\tilde{B}(f_s)] - \tilde{B}(f_0) = -D_{KL}(m \| \tilde{m}) \leq 0, \\ (\text{naive DM}) \quad & \mathbb{E}_{\tilde{m}}[\tilde{B}(f_s)] - \tilde{B}(f_0) = D_{KL}(\tilde{m} \| m) \geq 0, \end{aligned}$$

*with equality in either line if and only if  $m = \tilde{m}$  almost everywhere. Hence the sophisticated DM acquires the signal if and only if  $b \geq D_{KL}(m \| \tilde{m})$ , and the naive DM if and only if  $b \geq -D_{KL}(\tilde{m} \| m)$ .*

The psychological cost of information for a sophisticated DM is exactly  $C = D_{KL}(m \| \tilde{m})$ , that is, how well the signal is expected to discriminate between the objective and the motivated view of the world. The naive DM instead expects to gain  $D_{KL}(\tilde{m} \| m)$ , because he anticipates signals generated by his own motivated prior.

**Corollary 5** (Normal-linear case). *Under the assumptions of Corollary 1, with  $s = \omega + \epsilon$ ,  $\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$ , and  $\epsilon \perp \omega$ , we have  $m = \mathcal{N}(\mu_0, \sigma_0^2 + \sigma_\epsilon^2)$  and  $\tilde{m} = \mathcal{N}(\mu_0 + \alpha \sigma_0^2, \sigma_0^2 + \sigma_\epsilon^2)$ . Hence  $D_{KL}(m \| \tilde{m}) = D_{KL}(\tilde{m} \| m) = \frac{\alpha^2 \sigma_0^4}{2(\sigma_0^2 + \sigma_\epsilon^2)} = \frac{1}{2} \alpha^2 \lambda \sigma_0^2$ , which recovers Propositions 3 and 4; the gap between the two acquisition thresholds equals the symmetrized divergence  $\alpha^2 \lambda \sigma_0^2$ .*

**Theorem 3** (Blackwell monotonicity). *Let  $s'$  be generated from  $s$  through a state-independent stochastic kernel  $q(s' | s)$ , so that conditional on  $s$ , the distribution of  $s'$  does not depend on  $\omega$ . Then*

$$D_{KL}(m || \tilde{m}) \geq D_{KL}(m' || \tilde{m}') \quad \text{and} \quad D_{KL}(\tilde{m} || m) \geq D_{KL}(\tilde{m}' || m'),$$

where  $m'(s') = \int q(s' | s)m(s) ds$  and  $\tilde{m}'(s') = \int q(s' | s)\tilde{m}(s) ds$ . Consequently, a Blackwell more informative signal is weakly more psychologically costly for the sophisticated DM and weakly more psychologically attractive for the naive DM.

In the normal case,  $s' = s + \eta$  with independent Gaussian noise  $\eta$  is Blackwell less informative than  $s$  and has a lower  $\lambda$ . Theorem 3 therefore extends Hypothesis 1 to any pair of Blackwell-ranked signals.

**Theorem 4** (Stronger directional motives raise the cost of information). *Let  $v(\omega) = \alpha u(\omega)$  with  $\alpha \geq 0$  and  $u$  nondecreasing, let  $s \in \mathbb{R}$ , and suppose that, for every  $s' > s$ , the likelihood ratio  $p(s' | \omega)/p(s | \omega)$  is nondecreasing in  $\omega$ . Assume  $\int f_0(\omega) \exp(\alpha u(\omega)) d\omega < \infty$  on an open interval containing  $[0, \bar{\alpha}]$  and that differentiation under the integral sign is permitted. Then the sophisticated DM's psychological cost  $C(\alpha) = D_{KL}(m || \tilde{m}_\alpha)$  is nondecreasing in  $\alpha$  on  $[0, \bar{\alpha}]$ , with  $C(0) = 0$ .*

Theorem 4 extends Hypothesis 2 beyond the normal-linear case where  $C(\alpha) = \frac{1}{2}\alpha^2\lambda\sigma_0^2$  and the likelihood-ratio condition stated in Theorem 4 holds.

## C Proofs of Statements

Here we present the proofs for all statements in the paper.

**Proof of Remark 1.** A proof for this standard result can be found in, e.g., DeGroot (2005/1970).  $\square$

**Proof of Proposition 1.** The result is the continuous-state counterpart of Theorem 1 in Little (2025), where the optimization is performed over a finite-dimensional probability vector; Little (2025) states the continuous analogue as Theorem 5 in his Supplemental Information A.2. For completeness, we provide a proof in which the optimization is performed over a space of density functions. The proof proceeds in three steps.

*The problem.* First, let us restate the DM's optimization problem.

$$\tilde{f} = \arg \max_{f' \in \mathcal{F}} \{B(f, f')\} \quad (14)$$

$$= \arg \max_{f' \in \mathcal{F}} \{-D_{KL}(f' || f) + v(f')\} \quad (15)$$

$$= \arg \max_{f' \in \mathcal{F}} \left\{ - \int_{-\infty}^{\infty} f'(\omega) \log \left( \frac{f'(\omega)}{f(\omega)} \right) d\omega + \int_{-\infty}^{\infty} v(\omega) f'(\omega) d\omega \right\} \quad (16)$$

$$= \arg \max_{f' \in \mathcal{F}} \left\{ \int_{-\infty}^{\infty} f'(\omega) \left[ -\log \left( \frac{f'(\omega)}{f(\omega)} \right) + v(\omega) \right] d\omega \right\} \quad (17)$$

$$= \arg \max_{f' \in \mathcal{F}} \left\{ \int_{-\infty}^{\infty} f'(\omega) [v(\omega) + \log(f(\omega)) - \log(f'(\omega))] d\omega \right\} \quad (18)$$

$$= \arg \max_{f' \in \mathcal{F}} \left\{ \int_{-\infty}^{\infty} f'(\omega) [v(\omega) + \log(f(\omega))] d\omega - \int_{-\infty}^{\infty} f'(\omega) (\log f'(\omega)) d\omega \right\} \quad (19)$$

*Optimization.* Next, we introduce a Lagrange multiplier  $\eta$  to account for the constraint  $\int_{-\infty}^{\infty} f'(\omega) d\omega = 1$ , leading to the following Lagrangian:

$$\begin{aligned} \mathcal{L}(f', \eta) = & \int_{-\infty}^{\infty} f'(\omega) [v(\omega) + \log(f(\omega))] d\omega \\ & - \int_{-\infty}^{\infty} f'(\omega) \log(f'(\omega)) d\omega + \eta \left( 1 - \int_{-\infty}^{\infty} f'(\omega) d\omega \right) \end{aligned} \quad (20)$$

Taking the functional derivative with respect to  $f'(\omega)$  and setting it equal to zero gives:  $\frac{\delta \mathcal{L}}{\delta f'(\omega)} = v(\omega) + \log f(\omega) - \log f'(\omega) - 1 - \eta = 0$ . Simplifying it yields:  $\log f'(\omega) = v(\omega) + \log f(\omega) - \eta - 1$ . Exponentiation gives us:  $f'(\omega) = \exp(v(\omega) + \log f(\omega) - \eta - 1) = \exp(v(\omega)) \cdot \exp(-\eta - 1) \cdot f(\omega)$ . That is,  $f'(\omega) = f(\omega) \frac{\exp(v(\omega))}{\exp(1+\eta)}$ .

*Normalization.* To ensure that  $f'(\omega)$  is a valid probability density function, we impose  $\int_{-\infty}^{\infty} f'(\omega) d\omega = 1$ . Thus,  $1 = \int_{-\infty}^{\infty} f'(\omega) d\omega = \exp(-(1+\eta)) \int_{-\infty}^{\infty} f(\omega) \exp(v(\omega)) d\omega$ . Hence  $\exp(1+\eta) =$

$\int_{-\infty}^{\infty} f(\omega) \exp(v(\omega)) d\omega$ . Therefore, the optimal density function  $\tilde{f}(\omega)$  is:

$$\tilde{f}(\omega) = \frac{f(\omega) \exp(v(\omega))}{\int_{-\infty}^{\infty} f(\omega') \exp(v(\omega')) d\omega'} \quad (21)$$

Moreover, letting  $Z = \int_{-\infty}^{\infty} f(\omega) \exp(v(\omega)) d\omega$ , for every admissible density  $f'$ ,  $B(f, f') = \log Z - D_{KL}(f' \| \tilde{f}) \leq \log Z$ , with equality if and only if  $f' = \tilde{f}$  almost everywhere. Hence  $\tilde{f}$  is the unique global maximizer.  $\square$

**Proof of Corollary 1.** If  $v(\cdot)$  is linear in  $\omega$ , such that  $v(\omega) = \alpha\omega$ , then we can express  $v(\omega) = \alpha_0 + \alpha_1\omega + \alpha_2\omega^2$ , where  $\alpha_2 = 0$ . Following the logic of Corollary 6 in Little (2025), we prove that the density of the motivated belief must be proportional to:

$$\exp\left(\frac{-(\omega - \mu)^2}{2\sigma^2}\right) \cdot \exp(\alpha_2\omega^2 + \alpha_1\omega + \alpha_0) = \exp\left(\frac{-(\omega - \tilde{\mu})^2}{2\tilde{\sigma}^2}\right) \cdot \exp(k) \quad (22)$$

Taking logs of both sides gives

$$\frac{-(\omega - \mu)^2}{2\sigma^2} + \alpha_2\omega^2 + \alpha_1\omega + \alpha_0 = \frac{-(\omega - \tilde{\mu})^2}{2\tilde{\sigma}^2} + k \quad (23)$$

Expanding (23) and solving for  $\omega^2$ , we get  $-\frac{1}{2\sigma^2} + \alpha_2 = -\frac{1}{2\tilde{\sigma}^2}$  or

$$\tilde{\sigma}^2 = \frac{\sigma^2}{1 - 2\alpha_2\sigma^2} \quad (24)$$

Since  $\alpha_2 = 0$ ,  $\tilde{\sigma}^2|_{\alpha_2=0} = \sigma^2$ .

Expanding (23) and solving for  $\omega$ , we get  $\frac{\mu}{\sigma^2} + \alpha_1 = \frac{\tilde{\mu}}{\tilde{\sigma}^2}$  or

$$\tilde{\mu} = \frac{\mu + \alpha_1\sigma^2}{1 - 2\alpha_2\sigma^2} \quad (25)$$

Since  $\alpha_2 = 0$ ,  $\tilde{\mu}|_{\alpha_2=0} = \mu + \alpha_1\sigma^2$  where we equate  $\alpha_1 = \alpha$ ,  $\alpha \geq 0$ .  $\square$

**Proof of Proposition 2.** The proof is conceptually similar to Corollary 3 in Little (2025); however, here we extend it to a continuous-state case.

To do so, we need to have the objective function being evaluated at the optimal belief:

$$B(f, \tilde{f}) = -D_{KL}(\tilde{f}||f) + v(\tilde{f}) \quad (26)$$

$$= - \int_{-\infty}^{\infty} \tilde{f}(\omega) \log \left( \frac{\tilde{f}(\omega)}{f(\omega)} \right) d\omega + \int_{-\infty}^{\infty} v(\omega) \tilde{f}(\omega) d\omega \quad (27)$$

$$= - \int_{-\infty}^{\infty} \tilde{f}(\omega) \log \left( \frac{\exp(v(\omega))}{\int_{-\infty}^{\infty} f(\omega') \exp(v(\omega')) d\omega'} \right) d\omega + \int_{-\infty}^{\infty} v(\omega) \tilde{f}(\omega) d\omega \quad (28)$$

$$= - \int_{-\infty}^{\infty} \tilde{f}(\omega) \left[ \log(\exp(v(\omega))) - \log \left( \int_{-\infty}^{\infty} f(\omega') \exp(v(\omega')) d\omega' \right) \right] d\omega + \int_{-\infty}^{\infty} v(\omega) \tilde{f}(\omega) d\omega \quad (29)$$

$$= - \int_{-\infty}^{\infty} \tilde{f}(\omega) v(\omega) d\omega + \int_{-\infty}^{\infty} \tilde{f}(\omega) \log \left( \int_{-\infty}^{\infty} f(\omega') \exp(v(\omega')) d\omega' \right) d\omega + \int_{-\infty}^{\infty} v(\omega) \tilde{f}(\omega) d\omega \quad (30)$$

$$= \log \left( \int_{-\infty}^{\infty} f(\omega') \exp(v(\omega')) d\omega' \right) \int_{-\infty}^{\infty} \tilde{f}(\omega) d\omega \quad (31)$$

$$= \log \left( \int_{-\infty}^{\infty} f(\omega') \exp(v(\omega')) d\omega' \right) \quad (32)$$

That is, the objective function being evaluated at the optimal belief  $B(f, \tilde{f})$  is equal to

$$\log \left( \int_{-\infty}^{\infty} f(\omega') \exp(v(\omega')) d\omega' \right). \quad (33)$$

□

**Proof of Proposition 3.** Given that the sophisticated DM is aware of his directional motive and expects the posterior mean  $\mu_s$  to be equal to the prior mean  $\mu_0$ , the expected belief utility after acquiring the signal can be written as  $\mathbb{E}_s[\tilde{B}(f_s)] = \alpha\mu_0 + \frac{1}{2}\alpha^2(1-\lambda)\sigma_0^2$ . The utility from not acquiring the signal is simply the belief utility with the prior:  $\tilde{B}(f_0) = \alpha\mu_0 + \frac{1}{2}\alpha^2\sigma_0^2$ .

The DM will choose to acquire the signal if the utility with the signal and the overall benefit  $b$  exceeds the utility without the signal:  $\mathbb{E}_s[\tilde{B}(f_s)] + b \geq \tilde{B}(f_0)$ .

Substituting the respective expressions for  $\mathbb{E}_s[\tilde{B}(f_s)]$  and  $\tilde{B}(f_0)$ , we get:  $\alpha\mu_0 + \frac{1}{2}\alpha^2(1-\lambda)\sigma_0^2 + b \geq \alpha\mu_0 + \frac{1}{2}\alpha^2\sigma_0^2$ . Simplifying, this reduces to:  $b \geq \frac{1}{2}\alpha^2\lambda\sigma_0^2$ . For  $\alpha > 0$  and  $\lambda > 0$ ,  $\frac{1}{2}\alpha^2\lambda\sigma_0^2 > 0$ , so information is acquired only when the overall benefit  $b$  exceeds the psychological cost of information. □

**Proof of Proposition 4.** For the naive (unaware) motivated reasoner,  $\mathbb{E}_s[s] = \tilde{\mu}_0 = \mu_0 + \alpha\sigma_0^2$ . Hence,  $\mathbb{E}_s[\tilde{B}(f_s)] = \alpha(\lambda\tilde{\mu}_0 + (1-\lambda)\mu_0) + \frac{1}{2}\alpha^2(1-\lambda)\sigma_0^2 = \alpha\mu_0 + \frac{1}{2}\alpha^2(1+\lambda)\sigma_0^2$ . The utility from not acquiring the signal is:  $\tilde{B}(f_0) = \alpha\mu_0 + \frac{1}{2}\alpha^2\sigma_0^2$ . The DM acquires the signal iff:  $\mathbb{E}_s[\tilde{B}(f_s)] + b \geq \tilde{B}(f_0)$ , which is equivalent to:  $b \geq -\frac{1}{2}\alpha^2\lambda\sigma_0^2$ . □

**Proof of Proposition 5.** For the sophisticated (self-aware) asymmetry-sensitive motivated reasoner,  $\mathbb{E}_s[s] = \mu_0 - \delta$ . Hence,  $\mathbb{E}_s^S[\tilde{B}(f_s)] = \alpha(\lambda(\mu_0 - \delta) + (1 - \lambda)\mu_0) + \frac{1}{2}\alpha^2(1 - \lambda)\sigma_0^2 = \alpha\mu_0 + \frac{1}{2}\alpha^2(1 - \lambda)\sigma_0^2 - \alpha\lambda\delta$ . Since  $\tilde{B}(f_0) = \alpha\mu_0 + \frac{1}{2}\alpha^2\sigma_0^2$ , acquisition requires  $b \geq \frac{1}{2}\alpha^2\lambda\sigma_0^2 + \alpha\lambda\delta$ .

For the naive (unaware) asymmetry-sensitive motivated reasoner,  $\mathbb{E}_s[s] = \tilde{\mu}_0 - \delta = \mu_0 + \alpha\sigma_0^2 - \delta$ . Hence,  $\mathbb{E}_s^N[\tilde{B}(f_s)] = \alpha(\lambda(\tilde{\mu}_0 - \delta) + (1 - \lambda)\mu_0) + \frac{1}{2}\alpha^2(1 - \lambda)\sigma_0^2 = \alpha\mu_0 + \frac{1}{2}\alpha^2(1 + \lambda)\sigma_0^2 - \alpha\lambda\delta$ .

Thus acquisition requires:  $b \geq \alpha\lambda\delta - \frac{1}{2}\alpha^2\lambda\sigma_0^2$ .  $\square$

**Proof of Corollary 2.** Since  $b = \ell\lambda\sigma_0^2$ , Proposition 3 implies that the sophisticated DM acquires information if and only if  $\ell\lambda\sigma_0^2 \geq \frac{1}{2}\alpha^2\lambda\sigma_0^2$ , which simplifies to  $\ell \geq \frac{1}{2}\alpha^2$ .  $\square$

**Proof of Corollary 3.** Since  $b = \ell\lambda\sigma_0^2$ , Proposition 4 implies that information is acquired if and only if  $\ell\lambda\sigma_0^2 \geq -\frac{1}{2}\alpha^2\lambda\sigma_0^2$ . Since  $\lambda > 0$  and  $\sigma_0^2 > 0$ , this simplifies to  $\ell \geq -\frac{1}{2}\alpha^2$ . Since  $\ell > 0$ , this condition is always satisfied.  $\square$

**Proof of Corollary 4.** For the sophisticated (self-aware) asymmetry-sensitive motivated reasoner, Proposition 5 implies  $\ell\lambda\sigma_0^2 \geq \frac{1}{2}\alpha^2\lambda\sigma_0^2 + \alpha\lambda\delta$ . Dividing by  $\lambda\sigma_0^2$  yields  $\ell \geq \frac{1}{2}\alpha^2 + \alpha\delta\sigma_0^{-2}$ . For the naive (unaware) asymmetry-sensitive motivated reasoner, Proposition 5 implies  $\ell\lambda\sigma_0^2 \geq \alpha\lambda\delta - \frac{1}{2}\alpha^2\lambda\sigma_0^2$ . Dividing by  $\lambda\sigma_0^2$  yields  $\ell \geq \alpha\delta\sigma_0^{-2} - \frac{1}{2}\alpha^2$ .  $\square$

**Proof of Theorem 1.** Define  $g(s) \equiv \int f_s(\omega) \exp(v(\omega)) d\omega > 0$ . By Proposition 2,  $\tilde{B}(f_s) = \log g(s)$ .

For every  $\omega$ , Bayes' rule implies  $\int f_s(\omega) m(s) ds = \int p(s | \omega) f_0(\omega) ds = f_0(\omega)$ . Since the integrand below is nonnegative, Tonelli's theorem allows us to exchange the order of integration:

$$\begin{aligned} \mathbb{E}_m[g(s)] &= \int \int f_s(\omega) \exp(v(\omega)) d\omega m(s) ds \\ &= \int \exp(v(\omega)) \left( \int f_s(\omega) m(s) ds \right) d\omega \\ &= \int f_0(\omega) \exp(v(\omega)) d\omega = Z_0. \end{aligned}$$

Therefore, by the concavity of the logarithm,

$$\mathbb{E}_m[\tilde{B}(f_s)] = \mathbb{E}_m[\log g(s)] \leq \log \mathbb{E}_m[g(s)] = \log Z_0 = \tilde{B}(f_0).$$

Hence

$$C = \tilde{B}(f_0) - \mathbb{E}_m[\tilde{B}(f_s)] \geq 0.$$

Because the logarithm is strictly concave, equality holds if and only if  $g(s)$  is constant for  $m$ -almost every  $s$ . Finally, the acquisition condition in eq. (7) becomes  $b \geq C$ .  $\square$

**Proof of Theorem 2.** By Proposition 2,  $\tilde{B}(f_0) = \log Z_0$  and  $\tilde{B}(f_s) = \log g(s)$  and  $g(s) = \int f_s(\omega) \exp(v(\omega)) d\omega$ .

Using Bayes' rule,  $g(s) = \frac{1}{m(s)} \int p(s | \omega) f_0(\omega) \exp(v(\omega)) d\omega$ . Since  $f_0(\omega) \exp(v(\omega)) = Z_0 \tilde{f}_0(\omega)$ , we obtain  $g(s) = \frac{Z_0}{m(s)} \int p(s | \omega) \tilde{f}_0(\omega) d\omega = Z_0 \frac{\tilde{m}(s)}{m(s)}$ . Therefore,  $\tilde{B}(f_s) - \tilde{B}(f_0) = \log \left( \frac{\tilde{m}(s)}{m(s)} \right)$ .

Taking expectations under the objective predictive density  $m$  gives  $\mathbb{E}_m[\tilde{B}(f_s)] - \tilde{B}(f_0) = \mathbb{E}_m \left[ \log \left( \frac{\tilde{m}}{m} \right) \right] = -D_{KL}(m \| \tilde{m}) \leq 0$ . Taking expectations under the motivated predictive density  $\tilde{m}$  gives  $\mathbb{E}_{\tilde{m}}[\tilde{B}(f_s)] - \tilde{B}(f_0) = \mathbb{E}_{\tilde{m}} \left[ \log \left( \frac{\tilde{m}}{m} \right) \right] = D_{KL}(\tilde{m} \| m) \geq 0$ . By Gibbs' inequality, equality holds in either expression if and only if  $m = \tilde{m}$  almost everywhere. The stated acquisition conditions then follow directly from eq. (7).  $\square$

**Proof of Corollary 5.** By Corollary 1,  $\tilde{f}_0 = \mathcal{N}(\mu_0 + \alpha \sigma_0^2, \sigma_0^2)$ . Since  $s = \omega + \epsilon$ ,  $\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$ , and  $\epsilon \perp \omega$ , the objective and motivated predictive distributions are  $m = \mathcal{N}(\mu_0, \sigma_0^2 + \sigma_\epsilon^2)$  and  $\tilde{m} = \mathcal{N}(\mu_0 + \alpha \sigma_0^2, \sigma_0^2 + \sigma_\epsilon^2)$ . For two normal distributions with the same variance  $V$  and means differing by  $\Delta$ , the Kullback–Leibler divergence in either direction equals  $\frac{\Delta^2}{2V}$ . Here,  $\Delta = \alpha \sigma_0^2$  and  $V = \sigma_0^2 + \sigma_\epsilon^2$ . Hence

$$D_{KL}(m \| \tilde{m}) = D_{KL}(\tilde{m} \| m) = \frac{\alpha^2 \sigma_0^4}{2(\sigma_0^2 + \sigma_\epsilon^2)} = \frac{1}{2} \alpha^2 \lambda \sigma_0^2,$$

where  $\lambda = \frac{\sigma_0^2}{\sigma_0^2 + \sigma_\epsilon^2}$ . This reproduces the thresholds in Propositions 3 and 4.  $\square$

**Proof of Theorem 3.** Because the transformation from  $s$  to  $s'$  is governed by the same state-independent kernel  $q(s' | s)$  under both predictive distributions, the joint predictive densities of  $(s, s')$  are  $m(s)q(s' | s)$  and  $\tilde{m}(s)q(s' | s)$ . Their marginal densities for  $s'$  are  $m'$  and  $\tilde{m}'$ .

The chain rule for the Kullback–Leibler divergence gives

$$\begin{aligned} D_{KL}(m \parallel \tilde{m}) &= D_{KL}(mq \parallel \tilde{m}q) \\ &= D_{KL}(m' \parallel \tilde{m}') + \mathbb{E}_{m'} [D_{KL}(m(\cdot | s') \parallel \tilde{m}(\cdot | s'))] \\ &\geq D_{KL}(m' \parallel \tilde{m}'). \end{aligned}$$

The first equality holds because the same conditional density  $q(s' | s)$  appears under both distributions and therefore cancels from the likelihood ratio.

Reversing the roles of  $m$  and  $\tilde{m}$  gives  $D_{KL}(\tilde{m} \parallel m) \geq D_{KL}(\tilde{m}' \parallel m')$ . The result then follows from Theorem 2.  $\square$

**Proof of Theorem 4.** For each  $\alpha$ , define  $g_\alpha(s) = \int f_s(\omega) e^{\alpha u(\omega)} d\omega$  and  $\tilde{f}_{\alpha,s}(\omega) = \frac{f_s(\omega) e^{\alpha u(\omega)}}{g_\alpha(s)}$ . Also define  $h_\alpha(s) = \int u(\omega) \tilde{f}_{\alpha,s}(\omega) d\omega$ .

From the proof of Theorem 1,  $Z_0(\alpha) = \mathbb{E}_m[g_\alpha(s)]$ , and therefore  $C(\alpha) = \log Z_0(\alpha) - \mathbb{E}_m[\log g_\alpha(s)]$ . At  $\alpha = 0$ ,  $g_0(s) = 1$  for every  $s$ , so  $C(0) = 0$ .

Under the stated regularity conditions, differentiation gives  $\frac{\partial}{\partial \alpha} \log g_\alpha(s) = h_\alpha(s)$  and  $Z'_0(\alpha) = \mathbb{E}_m[g_\alpha(s) h_\alpha(s)]$ . Hence

$$C'(\alpha) = \frac{\mathbb{E}_m[g_\alpha(s) h_\alpha(s)]}{\mathbb{E}_m[g_\alpha(s)]} - \mathbb{E}_m[h_\alpha(s)] = \frac{\text{Cov}_m(g_\alpha(s), h_\alpha(s))}{\mathbb{E}_m[g_\alpha(s)]}$$

It remains to establish that both  $g_\alpha(s)$  and  $h_\alpha(s)$  are nondecreasing in  $s$ .

For  $s' > s$ , the assumption that  $\frac{p(s'|\omega)}{p(s|\omega)}$  is nondecreasing in  $\omega$  implies that  $\frac{f_{s'}(\omega)}{f_s(\omega)} = \frac{p(s'|\omega)}{p(s|\omega)} \frac{m(s)}{m(s')}$  is also nondecreasing in  $\omega$ . Thus, the posterior associated with the higher signal  $s'$  is larger in the likelihood-ratio order and therefore also in the first-order stochastic-dominance order.

Because  $u(\omega)$  is nondecreasing and  $\alpha \geq 0$ ,  $e^{\alpha u(\omega)}$  is nondecreasing in  $\omega$ . It follows that  $g_\alpha(s) = \mathbb{E}_{f_s}[e^{\alpha u(\omega)}]$  is nondecreasing in  $s$ .

Moreover,

$$\frac{\tilde{f}_{\alpha,s'}(\omega)}{\tilde{f}_{\alpha,s}(\omega)} = \frac{g_\alpha(s)}{g_\alpha(s')} \frac{f_{s'}(\omega)}{f_s(\omega)}.$$

The first factor does not depend on  $\omega$ , while the second is nondecreasing in  $\omega$ . Hence the motivated posteriors are also ordered by the likelihood ratio. Since  $u(\omega)$  is nondecreasing,  $h_\alpha(s) = \mathbb{E}_{\tilde{f}_{\alpha,s}}[u(\omega)]$  is therefore nondecreasing in  $s$ .

Thus  $g_\alpha(s)$  and  $h_\alpha(s)$  are two nondecreasing functions of the scalar signal  $s$ . Chebyshev's association inequality implies  $\text{Cov}_m(g_\alpha(s), h_\alpha(s)) \geq 0$ . Consequently,  $C'(\alpha) \geq 0$ , which proves that the sophisticated DM's psychological cost of information is nondecreasing in  $\alpha$ .  $\square$

**Proof of Theorem 2.** By Proposition 2,  $\tilde{B}(f_0) = \log Z_0$  and  $\tilde{B}(f_s) = \log g(s)$  with  $g(s) = \int f_s(\omega) \exp(v(\omega)) d\omega$ . Using Bayes' rule and  $f_0 \exp(v) = Z_0 \tilde{f}_0$ ,

$$g(s) = \frac{1}{m(s)} \int p(s | \omega) f_0(\omega) \exp(v(\omega)) d\omega = \frac{Z_0}{m(s)} \int p(s | \omega) \tilde{f}_0(\omega) d\omega = Z_0 \frac{\tilde{m}(s)}{m(s)}.$$

Therefore  $\tilde{B}(f_s) - \tilde{B}(f_0) = \log(\tilde{m}(s)/m(s))$ . Taking expectations under  $m$  gives  $\mathbb{E}_m[\log(\tilde{m}/m)] =$

$-D_{KL}(m||\tilde{m})$ ; taking them under  $\tilde{m}$  gives  $\mathbb{E}_{\tilde{m}}[\log(\tilde{m}/m)] = D_{KL}(\tilde{m}||m)$ . Non-negativity, and equality if and only if  $m = \tilde{m}$  almost everywhere, follow from Gibbs' inequality. The acquisition rules follow from eq. (7).  $\square$

**Proof of Corollary 5.** By Corollary 1,  $\tilde{f}_0 = \mathcal{N}(\mu_0 + \alpha\sigma_0^2, \sigma_0^2)$ , and adding independent noise  $\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$  yields the stated predictive densities. For two normal densities with common variance  $V$  and means differing by  $\Delta$ , both divergences equal  $\Delta^2/(2V)$ . Substituting  $\Delta = \alpha\sigma_0^2$ ,  $V = \sigma_0^2 + \sigma_\epsilon^2$  and  $\lambda = \sigma_0^2/(\sigma_0^2 + \sigma_\epsilon^2)$  gives the result.  $\square$

**Proof of Theorem 3.** Since  $q$  does not depend on  $\omega$ , the joint predictive densities of  $(s, s')$  are  $m(s)q(s' | s)$  and  $\tilde{m}(s)q(s' | s)$ , and their  $s'$ -marginals are  $m'$  and  $\tilde{m}'$ . The chain rule for the Kullback–Leibler divergence (Cover, 1999) gives

$$D_{KL}(m||\tilde{m}) = D_{KL}(m q || \tilde{m} q) = D_{KL}(m' || \tilde{m}') + \mathbb{E}_{m'}[D_{KL}(m(\cdot | s') || \tilde{m}(\cdot | s'))] \geq D_{KL}(m' || \tilde{m}'),$$

where the first equality holds because the common factor  $q$  cancels in the likelihood ratio, and  $m(\cdot | s')$ ,  $\tilde{m}(\cdot | s')$  are the conditional densities of  $s$  given  $s'$ . The argument for  $D_{KL}(\tilde{m}||m)$  is symmetric. The final statement follows from Theorem 2.  $\square$

**Proof of Theorem 4.** Write  $g_\alpha(s) = \int f_s(\omega)e^{au(\omega)}d\omega$ ,  $Z_0(\alpha) = \mathbb{E}_m[g_\alpha(s)]$  (see the proof of Theorem 1), and  $h_\alpha(s) = \int u(\omega)\tilde{f}_s(\omega)d\omega$ , where  $\tilde{f}_s \propto f_s e^{au}$  is the motivated posterior. Then  $C(\alpha) = \log Z_0(\alpha) - \mathbb{E}_m[\log g_\alpha(s)]$ , so  $C(0) = 0$  and

$$C'(\alpha) = \frac{\mathbb{E}_m[g_\alpha(s)h_\alpha(s)]}{\mathbb{E}_m[g_\alpha(s)]} - \mathbb{E}_m[h_\alpha(s)] = \frac{\text{Cov}_m(g_\alpha(s), h_\alpha(s))}{\mathbb{E}_m[g_\alpha(s)]},$$

using  $Z'_0(\alpha) = \int u f_0 e^{au} = \mathbb{E}_m[g_\alpha h_\alpha]$  by the same exchange of integrals as in Theorem 1. By MLRP,  $f_{s'}/f_s \propto p(s' | \cdot)/p(s | \cdot)$  is nondecreasing for  $s' > s$ , so the posteriors are ordered in the likelihood-ratio order and hence by first-order stochastic dominance. Since  $e^{au}$  is nondecreasing for  $\alpha \geq 0$ ,  $g_\alpha(s)$  is nondecreasing in  $s$ . Tilting by  $e^{au}$  multiplies  $f_{s'}/f_s$  by a positive constant, so the motivated posteriors inherit the likelihood-ratio order and  $h_\alpha(s)$  is also nondecreasing in  $s$ . By Chebyshev's association inequality, two nondecreasing functions of the scalar  $s$  have non-negative covariance; hence  $C'(\alpha) \geq 0$ .  $\square$